Equity and Capital Markets

Concept-First Notes + Exam Q&A + MCQ Bank + Mock Test — CA Intermediate

One complete file: concepts, worked Q&A, revision cheat-sheet, MCQs & a full mock test

Contents

Overview of Equity & Capital Markets

The Problem / Why this matters

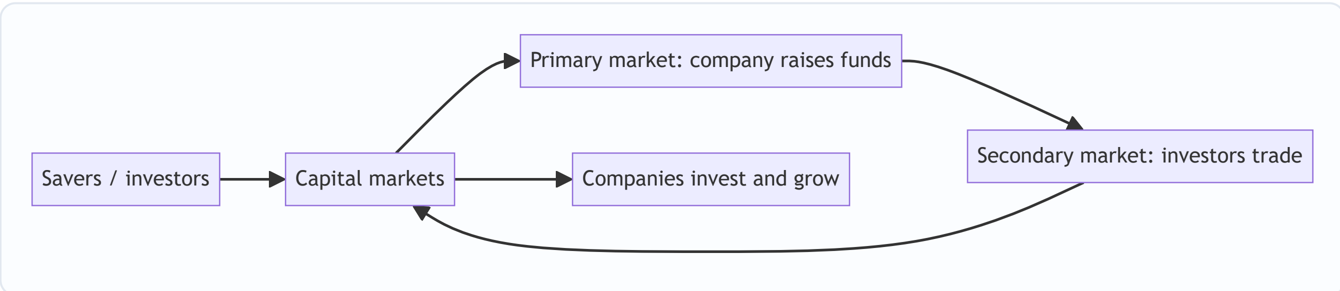
Companies need long-term capital to grow; savers need somewhere to put their money to earn a return. **Capital markets are the machinery that connects the two** — channelling household and institutional savings into productive businesses, and giving investors tradable claims on those businesses. Understanding how this system is structured — primary vs secondary, equity vs debt, the players and the plumbing — is the foundation for every equity, research, and markets role.

Core Idea

Capital markets are where **long-term funds** are raised and traded. The **primary market** is where new securities are issued and the company actually raises money; the **secondary market** is where those securities then trade among investors, providing the liquidity that makes the primary market work. Equity gives ownership; debt gives a creditor claim.

Why it works this way

Investors will only fund companies in the primary market if they believe they can **exit** later by selling to someone else — that exit is the secondary market. Liquidity in the secondary market lowers the return investors demand, which lowers the company's cost of capital. So the two markets are symbiotic: primary raises capital, secondary provides the liquidity that makes raising capital cheap.



Full technical content

Money markets vs capital markets: | | Money market | Capital market | |---|---|---| | Maturity | Short-term (< 1 yr) | Long-term (> 1 yr) | | Instruments | T-bills, CP, CDs, repo | Equities, bonds | | Purpose | Liquidity management | Long-term funding/investment |

Equity vs debt: | | Equity | Debt | |---|---|---| | Claim | Ownership, residual | Creditor, fixed | | Return | Dividends + capital gains, uncapped | Interest, capped | | Risk | Last in line, higher risk | Senior, lower risk | | Control | Voting rights | Usually none |

Primary vs secondary market: - **Primary** — first issuance: IPO, follow-on (FPO), rights issue, QIP, private placement, bond issuance. *The company receives the cash.* - **Secondary** — subsequent trading on an exchange between investors. *The company gets nothing; it provides price discovery and liquidity.*

Functions of capital markets: 1. **Capital formation** — channelling savings into investment. 2. **Price discovery** — continuous pricing of companies via trading. 3. **Liquidity** — the ability to convert holdings to cash. 4. **Risk transfer & sharing** — spreading risk across many investors. 5. **Corporate governance** — market discipline and monitoring by investors.

The ecosystem (previewing later chapters): issuers (companies), investors (retail, mutual funds, insurers, FPIs, pension funds), intermediaries (investment banks, brokers, exchanges, depositories, clearing houses), and the regulator (SEBI in India).

Worked examples

Example 1 — primary vs secondary cash flow. Company Z raises ₹1,000 cr in an IPO (primary) — that ₹1,000 cr goes to the company (or selling shareholders in an OFS). The next day, an investor buys ₹5 cr of Z shares on the NSE (secondary) — that ₹5 cr goes to the *seller*, not to Z. *Z only receives money at issuance.*

Example 2 — why liquidity lowers cost of capital. Two identical companies raise equity. Company A's shares will list on a liquid exchange (easy exit); Company B's won't trade at all. Investors demand a higher return from B for locking up their money, so B's cost of equity is higher and its shares are worth less. Liquidity is valuable.

Example 3 — equity vs debt in a good year and a bad year. A firm earns ₹100 cr. Bondholders get their fixed ₹20 cr interest either way. In a great year (₹200 cr), equity keeps the extra upside; in a terrible year (₹10 cr), equity is wiped out first while bondholders still rank ahead. Equity = uncapped upside, first-loss downside.

How it is tested in interviews

- **"Difference between primary and secondary markets?"** — "Primary is where the company issues new securities and actually raises money; secondary is where investors trade those securities among themselves, giving liquidity and price discovery. The company only receives cash in the primary market."
- **"When Infosys shares trade on the NSE, does Infosys get the money?"** — "No — that's the secondary market. Infosys only raised money at its IPO/issuances."
- **"Equity vs debt?"** — "Equity is ownership with uncapped, residual, higher-risk returns; debt is a senior, fixed, capped, lower-risk claim."
- **"What are the functions of capital markets?"** — Capital formation, price discovery, liquidity, risk sharing, governance.

Traps & common mistakes

- Thinking the company receives money from **secondary** trading.
- Confusing **money markets** (short-term) with **capital markets** (long-term).
- Treating equity and debt returns as symmetric — equity's downside is first-loss.
- Underrating the role of **liquidity** in lowering cost of capital.

First-principles recap

- Capital markets connect savers to companies needing long-term funds.
- **Primary** raises capital (company gets cash); **secondary** provides liquidity (investors trade).
- Equity = ownership/residual/uncapped; debt = senior/fixed/capped.

- Functions: capital formation, price discovery, liquidity, risk sharing, governance.
- Secondary-market liquidity lowers the primary-market cost of capital.

Quick-reference

Concept	One-liner
Primary market	New issue; company raises cash
Secondary market	Investors trade; liquidity & price discovery
Money vs capital market	Short-term vs long-term
Equity	Ownership, residual, uncapped
Debt	Creditor, fixed, senior

Primary Markets & the IPO Process

The Problem / Why this matters

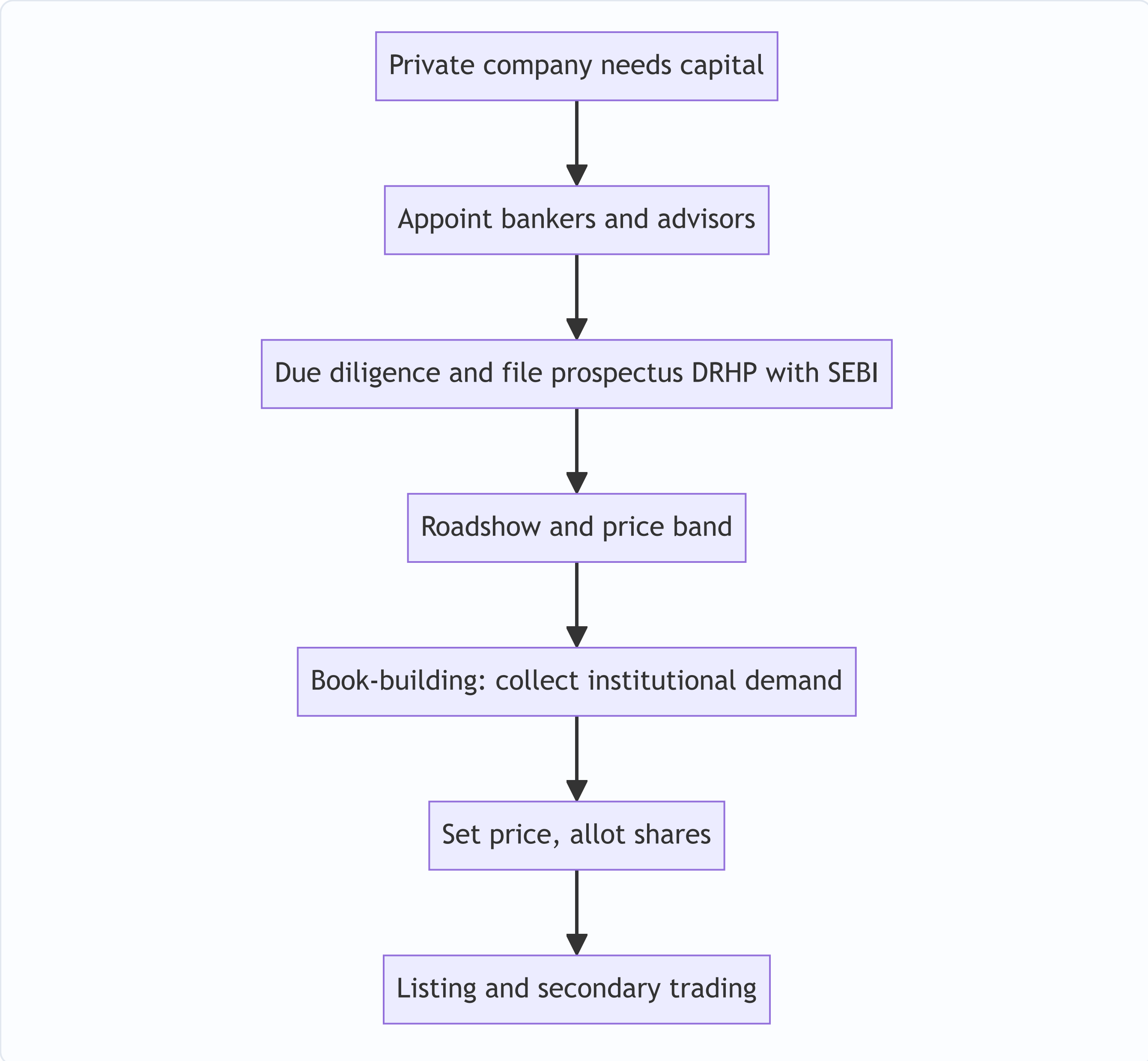
The most visible event in capital markets is a company **going public** — the IPO. It's how a private company raises large-scale equity, gives early investors an exit, and joins the public market. Understanding the IPO process — why companies list, how the price is set (book-building), who does what (the bankers, the regulator), and the risks (underpricing, lock-ups) — is essential for equity capital markets, investment banking, and research roles, and it comes up constantly in interviews.

Core Idea

The **primary market** is where companies issue securities to raise capital for the first time or afresh. An **IPO (Initial Public Offering)** is a private company's first sale of shares to the public, priced via **book-building**, intermediated by investment banks, and regulated (by SEBI in India). Beyond IPOs, primary issuance includes follow-ons, rights issues, QIPs and private placements.

Why it works this way

A company needs a mechanism to (i) discover a fair price when there's no market price yet, and (ii) distribute shares to many investors while managing risk. Book-building solves price discovery by collecting demand from institutions across a price band; underwriters manage distribution and (sometimes) guarantee the raise; the regulator protects investors through disclosure (the prospectus).



Full technical content

Why companies go public: raise growth capital, give founders/PE/VC an exit (offer for sale), acquire currency (listed shares) for M&A, raise profile/credibility, and enable employee stock liquidity. Costs: disclosure, scrutiny, short-term market pressure, listing expense, dilution.

Types of primary issuance: | Type | What it is | |---|---| | **IPO** | First public sale of shares | | **FPO / follow-on** | Further public issue by an already-listed firm | | **Rights issue** | Offer of new shares to existing holders, usually at a discount | | **QIP** | Qualified Institutional Placement — quick raise from institutions (India) | | **Private placement / preferential** | Shares sold to select investors | | **OFS** | Offer for Sale — existing holders sell (company gets no cash) |

Fresh issue vs offer for sale. A **fresh issue** creates new shares — the *company* receives the money (dilutes existing holders). An **OFS** sells *existing* shares — the selling shareholders receive the money (no dilution, no new company capital). Most IPOs are a mix.

The IPO process (India): 1. Appoint **merchant bankers (BRLMs)**, legal counsel, auditors, registrar. 2. Due diligence; draft the **DRHP** (Draft Red Herring Prospectus) and file with **SEBI**. 3. SEBI review and observations;

finalize the RHP. 4. **Roadshow** — market to institutions; set a **price band**. 5. **Book-building** — collect bids across the band from QIBs, non-institutional, and retail categories (India reserves quotas for each). 6. Determine the **cut-off price** from the demand book; allot shares. 7. **Listing** on NSE/BSE; secondary trading begins.

Pricing methods: book-building (price discovered from demand within a band — the norm) vs **fixed price** (set upfront). **Anchor investors** (large institutions) may be allotted a day before to signal quality.

Key phenomena: - **Underpricing** — IPOs often "pop" on day one (priced below where they trade), leaving money on the table but rewarding investors and ensuring a successful listing. - **Lock-up period** — insiders/pre-IPO holders can't sell for a set period after listing, preventing an immediate flood of stock. - **Greenshoe (over-allotment) option** — lets underwriters stabilize the price post-listing by selling up to ~15% extra and buying back if it falls. - **Winner's curse / oversubscription** — hot IPOs are heavily oversubscribed; retail allotment is scaled/lottery-based.

Worked examples

Example 1 — fresh issue vs OFS. An IPO raises ₹2,000 cr: ₹1,200 cr fresh issue (new shares → goes to the *company* for expansion) and ₹800 cr OFS (existing PE investor sells → goes to the *PE fund*, not the company). Only the ₹1,200 cr strengthens the company's balance sheet.

Example 2 — book-building price discovery. Price band ₹95–100. Institutional demand is strong at ₹100 (book covered 8× at the top), so the cut-off is set at **₹100**. Weak demand would have set it lower in the band. The market's bids, not the company's wish, set the price.

Example 3 — underpricing pop. Shares priced at ₹100 open at ₹135 on listing (+35%). Investors who got allotment gain; the company "left money on the table" (could have priced higher) but secured a strong, well-received listing and goodwill for future raises.

How it is tested in interviews

- **"Walk me through an IPO."** — Appoint bankers → due diligence and file DRHP with SEBI → roadshow and price band → book-building → set cut-off and allot → list and trade.
- **"Fresh issue vs OFS?"** — "Fresh issue creates new shares and the company gets the cash; OFS sells existing shares and the selling holders get the cash — no new capital for the company."
- **"What is book-building?"** — "Price discovery by collecting institutional demand across a price band and setting the cut-off from the demand book."
- **"Why are IPOs often underpriced?"** — "To ensure a successful, well-subscribed listing and reward investors for the uncertainty; it 'pops' but the issuer leaves some money on the table."
- **"What is a greenshoe / lock-up?"** — Greenshoe stabilizes the price via over-allotment; lock-up stops insiders dumping stock right after listing.

Traps & common mistakes

- Confusing **fresh issue** (company gets cash, dilution) with **OFS** (holders get cash, no dilution).
- Thinking the **company** sets the price — book-building lets **demand** set it.
- Forgetting **SEBI/DRHP** disclosure as the investor-protection backbone.
- Missing the purpose of **lock-ups and greenshoe** (orderly aftermarket).

First-principles recap

- Primary market = companies issue securities to raise capital.
- IPO = first public sale, priced by **book-building**, filed via **DRHP** with SEBI.
- **Fresh issue** funds the company; **OFS** cashes out existing holders.
- Underpricing, lock-ups, greenshoe, and anchor investors shape a successful listing.
- Beyond IPOs: FPO, rights, QIP, private placement.

Quick-reference

Term	Note
IPO	First public share sale
DRHP	Draft prospectus filed with SEBI
Book-building	Demand-based price discovery in a band
Fresh issue vs OFS	Company cash vs selling-holder cash
Greenshoe	Over-allotment price stabilization
Lock-up	Insiders can't sell for a set period

Secondary Markets & Trading Mechanics

The Problem / Why this matters

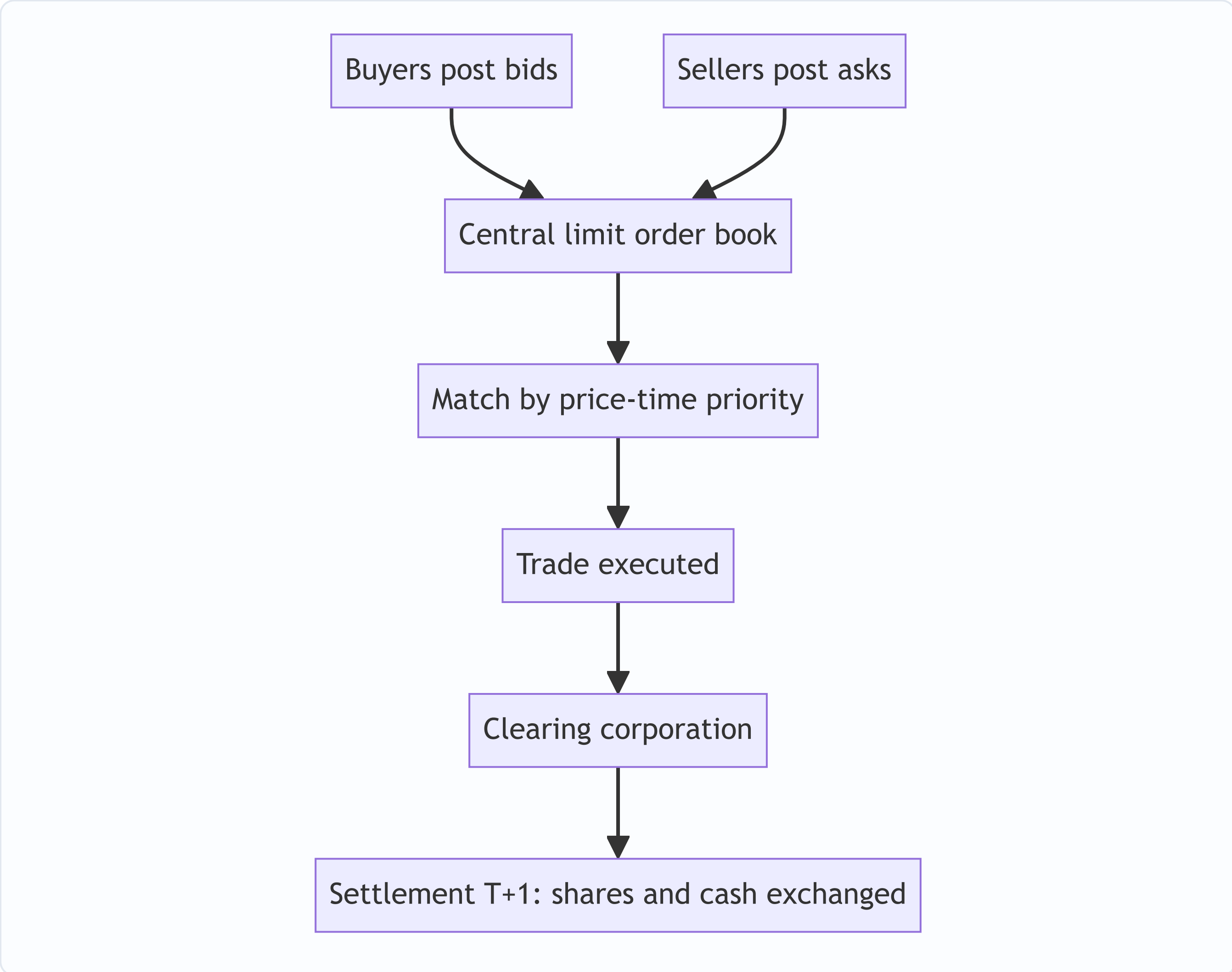
Once shares list, they trade — millions of times a day — on exchanges. How that trading actually works (order types, the order book, bid-ask spreads, matching, settlement) determines the price you get, the cost you pay, and the liquidity you rely on. Anyone touching markets — research, trading, operations, risk — must understand the plumbing. Interviewers test order types, the bid-ask spread, and settlement (T+1) directly.

Core Idea

The secondary market is a **continuous auction**: buyers post bids, sellers post asks, and the exchange's matching engine pairs them by price-time priority. The **bid-ask spread** is the cost of immediacy; **liquidity** is how easily you trade without moving the price; and every trade must **clear and settle** through depositories and clearing corporations.

Why it works this way

Price discovery needs a mechanism that continuously aggregates supply and demand. An **order-driven** market does this via a central limit order book where anyone can post orders and the best-priced ones execute first. Spreads exist because liquidity providers must be compensated for the risk of quoting; settlement infrastructure exists so buyers and sellers who never meet can trust that shares and cash actually change hands.



Full technical content

Market structures: order-driven (a central limit order book, as on NSE/BSE) vs **quote-driven** (dealers/market-makers quote two-way prices). Most modern equity markets are order-driven with electronic matching by **price-time priority** (best price first; at equal price, earliest order first).

The order book & bid-ask spread. The book shows all resting buy orders (bids) and sell orders (asks) at each price. The **best bid** and **best ask** define the spread; a **narrow spread = liquid**, a wide spread = illiquid and expensive to trade. **Depth** (quantity available near the top) shows how much you can trade without moving the price (**market impact**).

Order types: | Order | Behaviour | |---|---| | **Market** | Execute now at the best available price (certainty of execution, not price) | | **Limit** | Execute only at your price or better (certainty of price, not execution) | | **Stop-loss** | Becomes a market/limit order once a trigger price is hit | | **Stop-limit** | Stop that becomes a limit order | | Others | IOC, GTC, iceberg, day orders |

Liquidity providers. Market makers and high-frequency traders continuously quote both sides, earning the spread and supplying immediacy; they narrow spreads and add depth (see market microstructure).

Long vs short. Going **long** = buy, profit if price rises (max loss = amount invested). **Short selling** = sell borrowed shares to buy back cheaper; profit if price falls, but loss is theoretically unlimited (price can rise without bound). Shorting requires a stock-borrow and is regulated.

Clearing & settlement. After a trade: the **clearing corporation** (NSE Clearing / Indian Clearing) becomes central counterparty and nets obligations; **depositories** (NSDL, CDSL) hold shares in **demat** form and transfer them; cash and securities settle on **T+1** in India (among the fastest globally; India is moving toward T+0/instant). **Rolling settlement**, margining, and a settlement guarantee fund underpin trust.

Circuit breakers & price bands limit extreme moves (index-level halts and stock-level bands) to curb panic.

Worked examples

Example 1 — market vs limit order. Best bid ₹99.8, best ask ₹100.2 (spread 0.4). A **market buy** executes immediately at ₹100.2 (pays the spread for certainty). A **limit buy at ₹100.0** sits in the book and only fills if a seller drops to ₹100.0 — better price, but it might never execute. Choose by whether you value price or certainty.

Example 2 — spread and liquidity. Stock A (large-cap) shows spread ₹0.05 with 50,000 shares at the top — deep and liquid; you trade large size cheaply. Stock B (small-cap) shows spread ₹2.00 with 200 shares — illiquid; a modest order moves the price sharply (high market impact). Same rupee value order, very different execution cost.

Example 3 — settlement. You buy 100 shares on Monday (T). Under T+1, the shares are credited to your demat and cash debited on Tuesday (T+1). The clearing corporation guarantees the trade so you don't bear your counterparty's default risk.

How it is tested in interviews

- **"Market order vs limit order?"** — "Market executes now at the best price (certain execution, uncertain price); limit executes only at your price or better (certain price, uncertain execution)."
- **"What is the bid-ask spread and what does a wide spread mean?"** — "The gap between best bid and best ask — the cost of immediacy. A wide spread means illiquidity and expensive trading."
- **"What's the max loss on a short?"** — "Theoretically unlimited, because the price can rise without bound; a long's max loss is what you invested."
- **"How does settlement work in India?"** — "T+1 rolling settlement: shares (via NSDL/CDSL demat) and cash exchange one day after the trade, guaranteed by the clearing corporation as central counterparty."
- **"What does market depth tell you?"** — "How much you can trade near the top of the book without moving the price — your likely market impact."

Traps & common mistakes

- Confusing **market** (execution certainty) and **limit** (price certainty) orders.
- Forgetting a short's loss is **unlimited**.
- Ignoring **market impact/depth** — a big order in a thin book moves the price against you.
- Not knowing India settles at **T+1** (and is moving faster).
- Overlooking the **clearing corporation** as central counterparty removing settlement risk.

First-principles recap

- Secondary trading is a continuous auction matched by **price-time priority**.

- The **bid-ask spread** is the cost of immediacy; narrow = liquid.
- **Market** = certain execution; **limit** = certain price; **stop-loss** triggers on a level.
- Short selling has **unlimited** loss potential.
- Trades **clear and settle** via clearing corporations and depositories, T+1 in India.

Quick-reference

Item	Note
Order book	Resting bids/asks by price-time priority
Spread	Best ask − best bid (cost of immediacy)
Market order	Now, best price
Limit order	Your price or better
Short loss	Unlimited
Settlement	T+1 (NSDL/CDSL demat, clearing corp CCP)

Market Participants & Intermediaries

The Problem / Why this matters

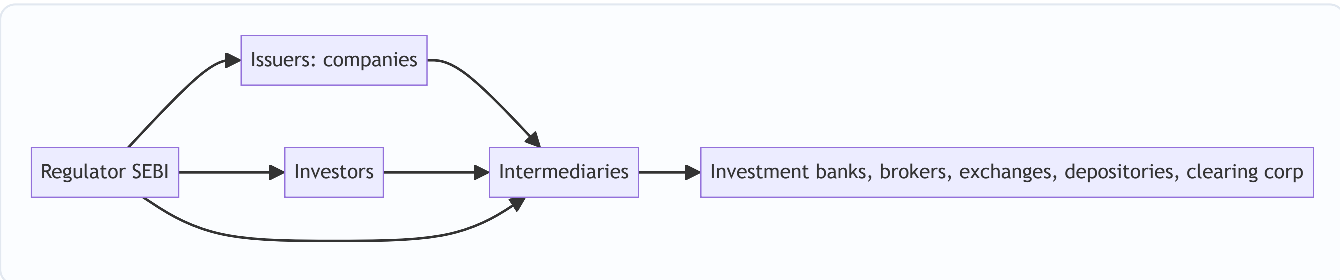
Markets don't run themselves — a web of participants and intermediaries makes issuance, trading, clearing and custody happen safely. Knowing **who does what** — issuers, the different investor types and their motivations, and the intermediaries (banks, brokers, exchanges, depositories, clearing corporations) — tells you how order flow moves, why prices react to certain investors, and where you'd fit in the ecosystem. Interviewers expect you to map the players cleanly.

Core Idea

The capital-market ecosystem has three groups: **issuers** who need capital, **investors** who supply it (each with different horizons and motivations), and **intermediaries** who connect and service them. Overlaying all of it is the **regulator** (SEBI) ensuring fairness, disclosure and investor protection.

Why it works this way

Issuers and investors can't transact directly at scale — they need someone to underwrite issues, match trades, hold securities, guarantee settlement and enforce rules. Each intermediary exists to solve a specific friction (distribution, execution, custody, counterparty risk, oversight), and the mix of investors determines how "smart," patient or reactive the money in a stock is.



Full technical content

Issuers: companies (private and public), governments (bonds), and other entities raising capital.

Investors (by type and motivation): | Investor | Horizon / motivation | |---|---| | **Retail** | Individuals; varied horizons; price-takers | | **Mutual funds / AMCs** | Pooled retail money; benchmark-driven | | **Insurance & pension funds** | Long-term, liability-matching, stable | | **FPIs / FIIs** | Foreign institutional flows; can be large and fast | | **Domestic institutions (DIIs)** | Banks, insurers, MFs domestically | | **Hedge funds / PMS / AIFs** | Absolute-return, flexible, can short/leverage | | **Promoters / strategic** | Control-oriented, long-term holders |

Institutional flows (FPI/DII) often drive index moves; "smart money" positioning is watched closely.

Intermediaries: | Intermediary | Role | |---|---| | **Investment bank / merchant banker (BRLM)** | Advise on and underwrite issues; M&A; capital raising | | **Brokers / trading members** | Execute investor orders on the exchange | | **Stock exchanges (NSE, BSE)** | Provide the trading platform and price discovery | | **Depositories (NSDL, CDSL)** | Hold securities in demat; effect transfer | | **Clearing corporations** | Central counterparty; net and guarantee settlement | | **Custodians** | Safekeep assets for institutions; settlement support | | **Registrars**

& Transfer Agents (RTAs) | Maintain shareholder records, process corporate actions | | **Rating agencies / research** | Independent credit/equity opinions | | **Investment advisers / distributors** | Advise and distribute to end investors |

The regulator — SEBI. Sets rules for issuers (disclosure), intermediaries (registration, conduct), and markets (surveillance, insider-trading and manipulation enforcement); protects investors and develops the market. RBI oversees banking/monetary aspects; IRDAI insurance.

Sell-side vs buy-side (preview). *Sell-side* (banks, brokers) creates products, executes and publishes research to win client business. *Buy-side* (asset managers, funds) invests client money and consumes that research to make decisions.

Worked examples

Example 1 — the chain of a single trade. A mutual fund (investor) instructs its **broker** (intermediary) to buy 1 lakh shares on the **NSE** (exchange); the trade matches, the **clearing corporation** guarantees it, the **depository (CDSL/NSDL)** moves the shares into the fund's **custodian's** demat account on T+1. Five different intermediaries touched one trade, each solving a distinct friction.

Example 2 — why the investor mix matters. Two similar mid-caps: Stock A is 70% held by long-term insurers and index funds (stable, patient); Stock B is 40% held by fast FPIs and traders (volatile, flow-driven). A negative headline moves B far more, because its holder base reacts and rotates quickly. *Knowing the holder base predicts volatility.*

Example 3 — SEBI's role. A promoter tries to sell shares on undisclosed bad news (insider trading). SEBI's surveillance flags the trades, investigates, and penalizes — protecting other investors and preserving trust that keeps the market liquid.

How it is tested in interviews

- **"Who are the main market participants?"** — Issuers; investors (retail, mutual funds, insurers/pensions, FPIs, DIIs, hedge funds/PMS, promoters); intermediaries (banks, brokers, exchanges, depositories, clearing corporations, custodians, RTAs); regulator (SEBI).
- **"What does a depository do?"** — "Holds securities in dematerialized form and effects their transfer — NSDL and CDSL in India."
- **"What's the difference between sell-side and buy-side?"** — "Sell-side creates, executes and researches to serve clients; buy-side invests client money and consumes that research."
- **"Why do FPI flows matter?"** — "They're large and can move quickly, so they often drive index-level moves and volatility."
- **"What does SEBI do?"** — "Regulates issuers, intermediaries and markets — disclosure, conduct, surveillance, and investor protection."

Traps & common mistakes

- Confusing **depository** (holds shares) with **clearing corporation** (guarantees settlement) with **exchange** (matches trades) — three different roles.
- Lumping all investors together — horizons and behaviour differ sharply.
- Mixing up **sell-side** and **buy-side**.

- Forgetting the **regulator** as the backbone of trust.

First-principles recap

- Three groups: issuers, investors, intermediaries — plus the regulator.
- Each intermediary solves a friction: distribution, execution, custody, settlement, records.
- The **investor mix** shapes a stock's volatility and behaviour.
- Exchange (match) ≠ clearing corp (guarantee) ≠ depository (hold).
- SEBI ensures disclosure, conduct and investor protection.

Quick-reference

Player	Role
Investment bank / BRLM	Underwrite/advise issues
Broker	Execute orders
Exchange (NSE/BSE)	Match trades, price discovery
Depository (NSDL/CDSL)	Hold & transfer demat shares
Clearing corporation	CCP, guarantee settlement
SEBI	Regulate & protect investors

Equity Instruments

The Problem / Why this matters

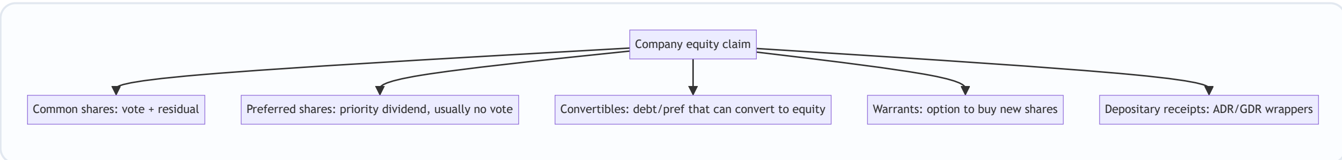
"Equity" isn't a single thing. Companies issue common shares, preferred shares, depositary receipts, warrants, and convertibles — each with different rights to cash flow, control and risk. Knowing the instruments, their rights, and where they sit in the capital structure lets you understand what you actually own, how it's valued, and why one class trades differently from another. It's foundational for equity research and capital markets roles.

Core Idea

Equity instruments are claims on a company's **residual value and control**, ranging from plain common shares (full ownership and residual risk) through preferred shares (a hybrid with priority) to derivative-like claims (warrants, convertibles) and cross-border wrappers (ADRs/GDRs). Each differs in cash-flow priority, voting rights, and payoff.

Why it works this way

Different investors want different risk-return-control packages, and companies want to tailor securities to tap the widest, cheapest capital. So the market slices the equity claim into instruments: some trade voting rights for a fixed dividend (preferred), some bundle upside options (warrants, convertibles), and some repackage foreign shares for local investors (depositary receipts).



Full technical content

Common (equity/ordinary) shares: full ownership — voting rights, residual claim on profits (dividends, discretionary) and on assets in liquidation (last in line). Uncapped upside, first-loss downside. The default meaning of "a share."

Preferred shares: a hybrid. Priority over common for **dividends** (often fixed) and in **liquidation**, but usually **no vote**. Variants: **cumulative** (missed dividends accrue), **participating** (share in extra upside), **convertible** (can convert to common), **redeemable/callable**. Sits between debt and common equity in the capital structure.

Depositary receipts: let investors hold foreign shares locally. - **ADR** (American Depositary Receipt) — a US-listed certificate representing shares of a non-US company (e.g., Infosys ADR on NYSE). - **GDR** (Global Depositary Receipt) — similar, listed outside the home and US markets. - Purpose: access foreign capital and investors without a full foreign listing.

Warrants: a company-issued option giving the holder the right to buy **new** shares at a set price for a period. Exercise creates new shares (dilutive). Often attached to bonds as a sweetener.

Convertible securities: bonds or preferred shares that can convert into common equity at a set ratio. Give the holder debt-like downside protection plus equity upside; give the issuer a lower coupon. Convert when the equity is worth more than the bond.

Rights & entitlements: existing shareholders may receive **rights** (to buy new shares at a discount — see corporate actions) or **bonus** shares.

Key rights of a shareholder: vote (elect directors, major decisions), dividends (if declared), residual assets in liquidation, information/disclosure, pre-emption (rights issues), and to transfer shares.

Where each sits (priority in liquidation): secured debt → unsecured debt → **preferred equity** → **common equity** (last). Higher priority = lower risk = lower expected return.

Worked examples

Example 1 — common vs preferred in a bad year. A firm can pay ₹10 cr of dividends. Preferred holders are owed a fixed ₹6 cr (cumulative) — they're paid first; common gets the remaining ₹4 cr (discretionary). In a terrible year with ₹4 cr available, preferred takes it all and common gets nothing (and cumulative preferred arrears accrue). Preferred = priority but capped; common = residual and variable.

Example 2 — convertible economics. A ₹1,000 convertible bond converts into 8 shares (conversion price ₹125). If the stock is ₹100, the holder keeps the bond (worth more than 8 × ₹100 = ₹800) and enjoys downside protection. If the stock rises to ₹180, converting gives 8 × ₹180 = ₹1,440 — the holder converts and captures the equity upside. The issuer paid a low coupon for embedding that option.

Example 3 — ADR arbitrage link. Infosys trades in Mumbai and as an ADR in New York. If the ADR (adjusted for the ratio and FX) diverges from the Mumbai price, arbitrageurs trade both until they realign — so the ADR essentially tracks the home share.

How it is tested in interviews

- **"Common vs preferred shares?"** — "Common has voting rights and a residual, variable claim (uncapped upside, first-loss). Preferred usually has no vote but priority on a fixed dividend and in liquidation — a hybrid between debt and common."
- **"What is an ADR?"** — "A US-listed certificate representing shares of a foreign company, letting US investors hold it in dollars without a foreign listing."
- **"How does a convertible bond work?"** — "A bond that can convert into equity at a set ratio; the holder gets debt downside protection plus equity upside, and the issuer pays a lower coupon for that option."
- **"Where does preferred sit in the capital structure?"** — "Below debt, above common — priority over common for dividends and liquidation, but behind all creditors."

Traps & common mistakes

- Thinking preferred is "safer equity" without noting it's still **below all debt**.
- Confusing **warrants** (company issues new shares, dilutive) with exchange-traded **options** (no new shares).
- Forgetting convertibles are **dilutive** on conversion.
- Treating ADRs/GDRs as different companies rather than **wrappers** on the same shares.

First-principles recap

- Equity instruments slice the ownership claim by cash-flow priority, control and payoff.
- **Common** = vote + residual (uncapped/first-loss); **preferred** = priority dividend, usually no vote.
- **Convertibles** = debt/pref with an equity option; **warrants** = right to buy new shares (dilutive).
- **ADR/GDR** = wrappers letting foreign investors hold local shares.
- Priority: debt → preferred → common.

Quick-reference

Instrument	Key feature
Common share	Vote + residual claim
Preferred share	Priority dividend, usually no vote
Convertible	Converts debt/pref → equity
Warrant	Right to buy new shares (dilutive)
ADR / GDR	Foreign-share wrapper
Priority	Debt > preferred > common

Market Indices

The Problem / Why this matters

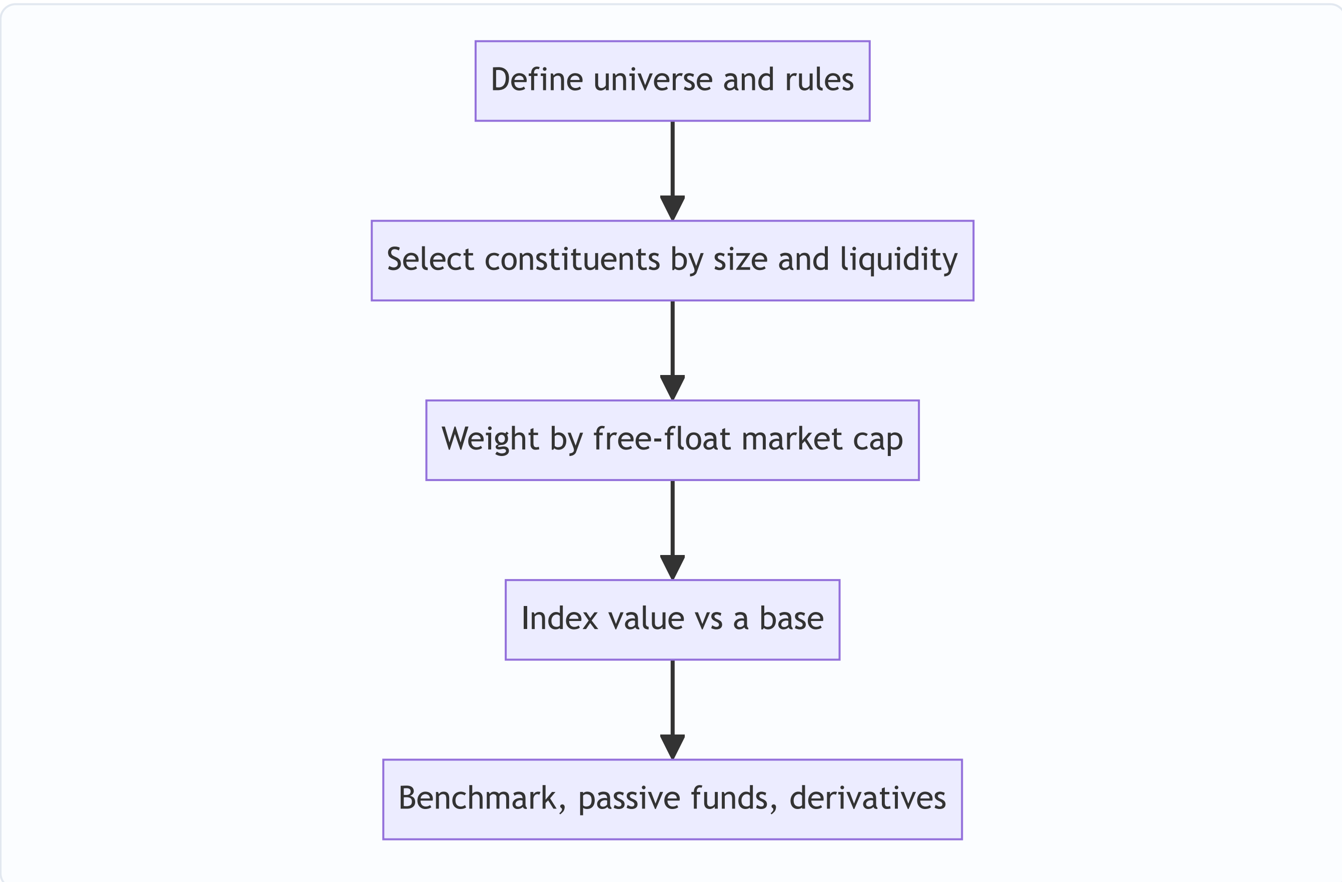
"The market was up 1% today" — but what *is* "the market"? An **index** is the number that represents it. Indices are how performance is measured, how passive funds are built, how benchmarks are set, and how derivatives (index futures/options) are priced. Understanding how they're constructed (and their quirks) matters for research, portfolio, and derivatives roles — and interviewers ask how the Nifty/Sensex are built and what "free-float market-cap weighted" means.

Core Idea

A market index is a **single number tracking the value of a defined basket of stocks**, used as a barometer of the market or a segment. Most modern indices are **free-float market-capitalization weighted**: each stock's weight is proportional to the market value of its publicly tradable shares.

Why it works this way

Investors need a comparable, investable benchmark. Weighting by free-float market cap makes the index reflect where investable money actually sits (bigger, more freely-traded companies matter more) and makes it **replicable** by index funds. It also self-adjusts: as a stock rises, its weight rises, mirroring a real buy-and-hold portfolio.



Full technical content

Construction methods (weighting): | Method | Weight by | Examples | |---|---|---| | **Free-float market-cap** | Value of publicly tradable shares | Nifty 50, Sensex, S&P 500 | | Full market-cap | Total shares × price | older indices | | **Price-weighted** | Share price (not size) | Dow Jones, Nikkei | | Equal-weighted | Same weight each | equal-weight indices |

Free-float excludes promoter/strategic/locked holdings — only shares actually available to trade count, so the index reflects investable value.

Indian benchmarks: - **Nifty 50** (NSE) — 50 large-caps, free-float market-cap weighted. - **Sensex** (BSE) — 30 large-caps, free-float weighted. - Broader: Nifty Next 50, Nifty 100/500, Nifty Midcap/Smallcap, and sectoral indices (Bank Nifty, Nifty IT, etc.).

The divisor & continuity. An index uses a **divisor** so that mechanical changes (constituent additions/deletions, corporate actions, new share issuance) don't create artificial jumps — the divisor is adjusted to keep the index continuous. This is why a stock split doesn't move the index.

Rebalancing. Indices are reviewed periodically (e.g., semi-annually for Nifty) to add/drop constituents by size and liquidity rules. Index inclusion/exclusion drives real flows (passive funds must buy the added stock), so it moves prices.

Uses of indices: 1. **Benchmark** — measure a portfolio's relative performance (alpha). 2. **Passive investing** — index funds and ETFs replicate the index cheaply. 3. **Derivatives** — index futures and options (Nifty/Bank Nifty are among the world's most traded). 4. **Market sentiment** — a barometer of the economy/segment.

Price return vs total return index. A **price index** ignores dividends; a **total return index (TRI)** reinvests dividends — TRI is the fair benchmark for a fund that also earns dividends.

Worked examples

Example 1 — free-float weighting. Company X has ₹1,000 cr total market cap but the promoter holds 60% (locked), so free-float = 40% = ₹400 cr. Company Y has ₹600 cr market cap, 100% free-float = ₹600 cr. Despite X being "bigger," Y gets a larger index weight (₹600 cr vs ₹400 cr of investable value). *Free-float, not total size, drives weight.*

Example 2 — why a split doesn't move the index. A constituent does a 1:1 split — price halves, shares double, market cap unchanged. The index divisor absorbs the mechanical change so the index value is unaffected. Corporate actions don't distort the index.

Example 3 — index inclusion flow. A stock is added to the Nifty 50 at the next rebalance. Every passive fund tracking the Nifty must buy it to match the index, creating real buying pressure — the stock often rises into and on inclusion. Index membership has real cash-flow consequences.

How it is tested in interviews

- **"How is the Nifty/Sensex constructed?"** — "Free-float market-capitalization weighted baskets of large-caps (Nifty 50 on NSE, Sensex 30 on BSE); weights reflect the value of publicly tradable shares."
- **"What does free-float mean and why use it?"** — "Only publicly tradable shares (excluding promoter/locked holdings). It makes the index reflect investable value and be replicable by index funds."

- **"Why doesn't a stock split move the index?"** — "The divisor adjusts for mechanical changes so the index stays continuous; market cap is unchanged by a split."
- **"Price-weighted vs market-cap-weighted?"** — "Price-weighted (Dow) weights by share price regardless of size; market-cap weighted (Nifty/S&P) weights by company value — the latter reflects the real market."
- **"Price return vs total return index?"** — "TRI reinvests dividends; it's the fair benchmark for a fund that also collects dividends."

Traps & common mistakes

- Saying "market-cap weighted" without the **free-float** nuance for Nifty/Sensex.
- Thinking a **split or bonus** moves the index (the divisor absorbs it).
- Confusing **price-weighted** (Dow) with **market-cap-weighted** (S&P/Nifty).
- Benchmarking a fund against a **price index** instead of the **total-return** index.

First-principles recap

- An index = a single number tracking a defined basket; the market's barometer.
- Nifty/Sensex are **free-float market-cap weighted** — investable value drives weight.
- A **divisor** keeps the index continuous through corporate actions and rebalances.
- Indices power benchmarking, passive funds, and derivatives.
- Use the **total-return** index to fairly benchmark dividend-earning funds.

Quick-reference

Item	Note
Nifty 50 / Sensex	Free-float market-cap weighted (50 / 30 stocks)
Free-float	Publicly tradable shares only
Divisor	Keeps index continuous through changes
Price-weighted	Dow, Nikkei (by share price)
TRI	Reinvests dividends (fair benchmark)
Inclusion	Forces passive-fund buying (real flow)

The Equity Research Process

The Problem / Why this matters

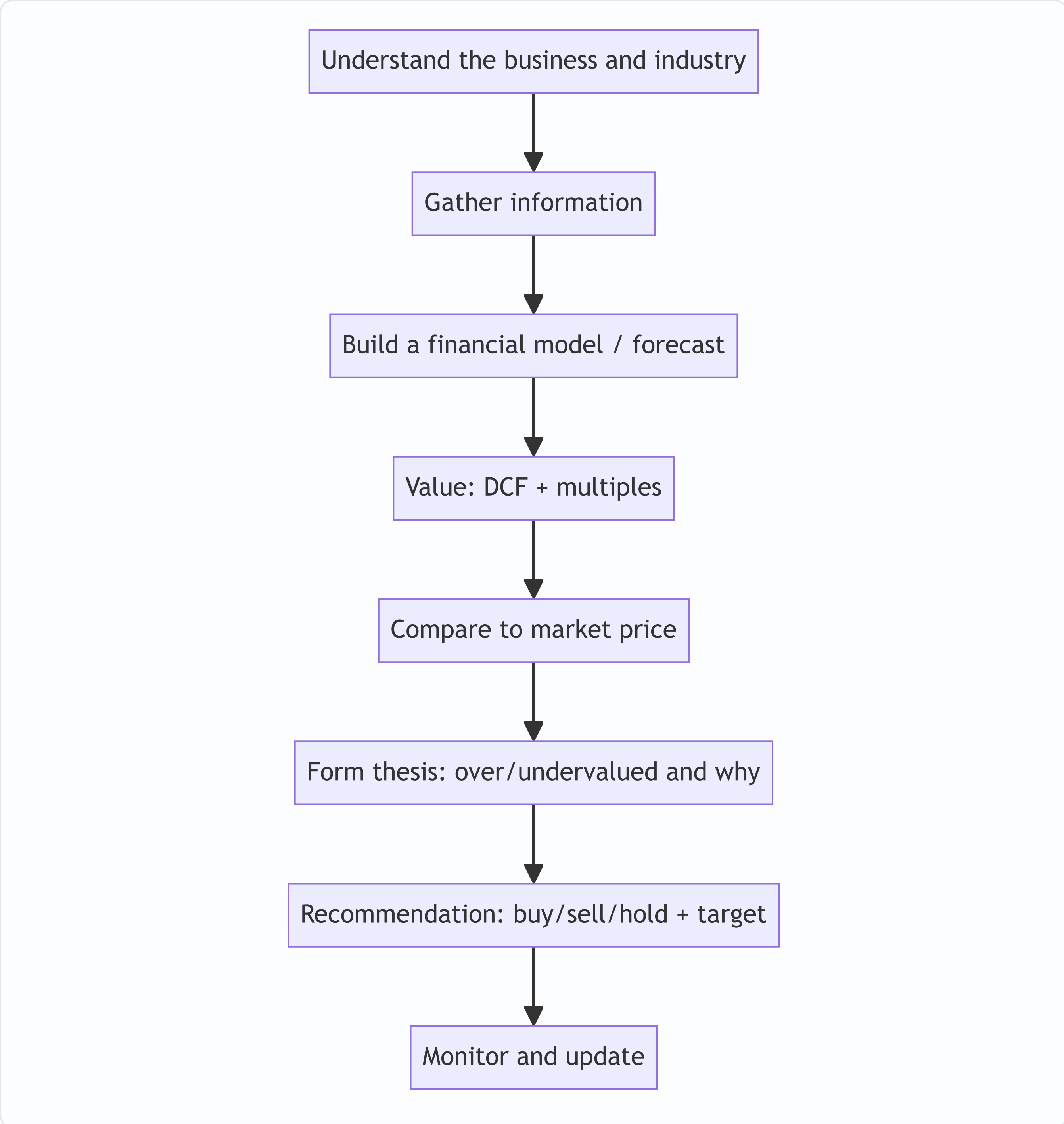
An equity research analyst's job is to form and defend a **view on a stock** — is it worth more or less than the market says, and why? That view drives buy/sell/hold recommendations that clients act on. Knowing the end-to-end process — coverage, information gathering, modelling, valuation, thesis, and the recommendation — is exactly what an equity research or investment-analyst interview tests, often as "walk me through how you'd analyse a company."

Core Idea

Equity research is a repeatable process: **understand the business** → **model its financials** → **value it** → **compare to the market price** → **form a thesis and recommendation** → **monitor.** The output is an investment view supported by a model and a written note.

Why it works this way

Markets are mostly efficient, so to add value an analyst must either know something the market underappreciates or interpret known facts better. That requires deeply understanding the business and its drivers, building a defensible forecast, and translating it into a value that can be compared to the price — then articulating *why* the market is wrong.



Full technical content

Step 1 — Understand the business & industry. What does the company do, how does it make money (revenue model, unit economics), what's its competitive position and moat, who are its customers and competitors, and what drives the industry (see fundamental analysis).

Step 2 — Gather information. Annual reports/10-Ks, quarterly results and concalls, management guidance, industry data, channel checks, expert networks, competitor filings, and news. Rigorous, primary-source-driven.

Step 3 — Build the model. A three-statement model with a **driver-based forecast** (revenue drivers → costs → the three linked statements), producing the free cash flows and metrics valuation needs (see the modeling chapter).

Step 4 — Value. Intrinsic (DCF) and relative (comps/multiples), triangulated to a value range and a target price (see applied valuation).

Step 5 — Form the thesis. The core argument: *why* the stock is mispriced. A good thesis is **differentiated** (a view the market doesn't fully hold), **specific**, and **falsifiable** (you can say what would prove you wrong). Identify the **catalysts** that will close the value gap and the **risks**.

Step 6 — Recommendation. Buy / Hold / Sell (or Overweight/Neutral/Underweight) with a **target price** and time horizon, and the expected return vs the current price.

Step 7 — Write and communicate. The research note (initiation or update) and verbal pitch to clients/PMs; then **monitor** results, news and thesis, updating the view as facts change.

The analyst's edge — three sources of alpha: | Edge | Description | |---|---| | **Informational** | Knowing a fact the market hasn't priced (rare, and insider info is illegal) | | **Analytical** | Interpreting known facts better (better model/insight) | | **Behavioural/time-horizon** | Exploiting others' biases or short-termism |

Most legitimate edge is **analytical** and **behavioural**.

Quality markers of good research: a clear, differentiated thesis; a defensible model with sensible assumptions; explicit catalysts and risks; and intellectual honesty (state what would make you wrong).

Worked examples

Example 1 — the process end to end. Analyst covers an FMCG firm. Understands the business (rural distribution moat); models revenue off volume × price with margin expansion from premiumization; DCFs it and cross-checks EV/EBITDA vs peers → intrinsic ₹1,200 vs market ₹1,000. Thesis: the market underrates margin expansion from premiumization (differentiated view); catalyst: next two quarters of margin beats; risk: rural slowdown. Recommendation: **Buy, target ₹1,200, +20%.**

Example 2 — differentiated vs consensus thesis. Two analysts value the same stock at ₹1,000, exactly the market price, agreeing with consensus. Neither adds value — a research view is only useful when it *differs* from the price and has a reason. The job is to find and defend the gap.

Example 3 — thesis falsification. An analyst's Buy rests on a new product driving 15% volume growth. She states: "If the first two quarters show under 8% volume growth, the thesis is broken and I'll downgrade." That falsifiability is what separates research from cheerleading.

How it is tested in interviews

- **"Walk me through how you'd analyse a company / research a stock."** — Recite: understand the business → gather info → model → value (DCF + comps) → compare to price → form a differentiated thesis with catalysts and risks → recommend with a target → monitor.
- **"What makes a good investment thesis?"** — "Differentiated from consensus, specific, with clear catalysts and risks, and falsifiable — you can state what would prove you wrong."
- **"Where does an analyst's edge come from?"** — "Mostly analytical (better interpretation) and behavioural (exploiting short-termism/biases); informational edge is rare and insider info is illegal."
- **"How do you decide buy vs sell?"** — "Compare intrinsic value/target to the market price; the gap and catalysts drive the rating and expected return."

Traps & common mistakes

- Producing a view that **matches consensus/price** — it adds no value.
- A model with no **thesis** — data without a differentiated argument.
- No **catalysts** (why does the gap close?) or **risks** (what if you're wrong?).
- Not stating what would **falsify** the thesis — a recipe for holding losers.
- Chasing **informational edge** into insider-trading territory.

First-principles recap

- Research = understand → model → value → compare to price → thesis → recommend → monitor.
- Value is only added when your view **differs from the market**, with a reason.
- A good thesis is differentiated, specific, catalyst-driven, and **falsifiable**.
- Edge is mostly analytical and behavioural.
- The output is a rating, a target price, and a defensible note.

Quick-reference

Step	Output
Understand business	Revenue model, moat, drivers
Model	Driver-based 3-statement forecast
Value	DCF + multiples → target
Thesis	Why mispriced + catalysts + risks
Recommendation	Buy/Hold/Sell + target + horizon
Edge	Analytical + behavioural

Fundamental Analysis: Top-Down & Bottom-Up

The Problem / Why this matters

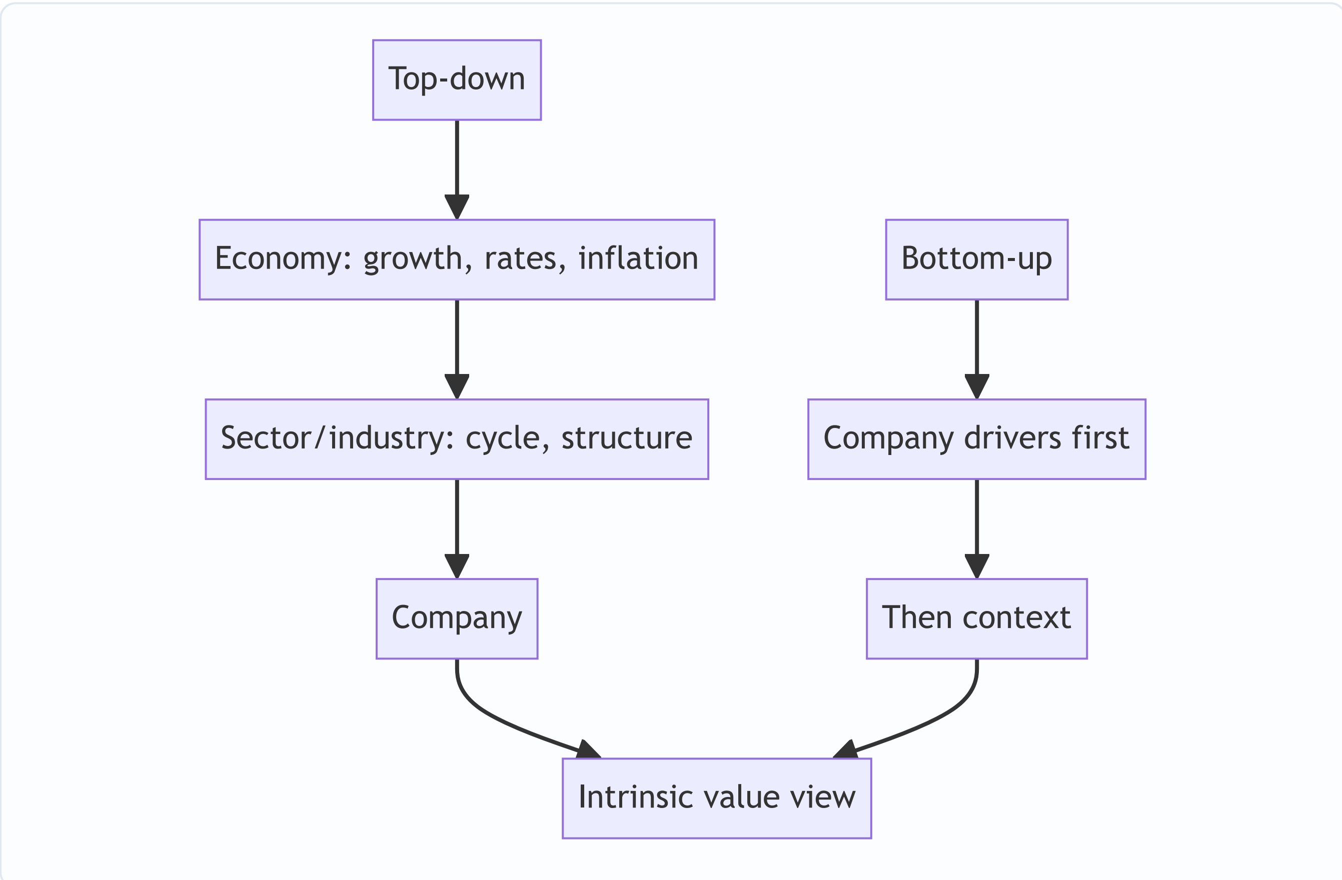
To decide whether a stock is cheap or dear, you must first understand the *fundamentals* — the real economic drivers of the business and its environment. **Fundamental analysis** is the discipline of assessing a company's intrinsic worth from the economy, industry and company up (or down). It's the core skill of equity research and long-term investing, and interviewers probe whether you can structure an analysis rather than just quote a P/E.

Core Idea

Fundamental analysis estimates a company's intrinsic value from its underlying economics. It's done **top-down** (economy → sector → company) and **bottom-up** (company-specific analysis first), usually combining both: understand the macro and industry context, then dig into the specific company's drivers, quality and valuation.

Why it works this way

A company's results are shaped by forces at three levels — the macro economy (growth, rates, inflation), the industry (structure, cycle, competition), and the company itself (strategy, execution, moat). Ignoring any level gives a wrong answer: a great company in a collapsing industry, or a mediocre one riding a boom, will disappoint if you only looked at one layer.



Full technical content

Top-down analysis (economy → sector → stock): 1. **Macro** — GDP growth, interest rates, inflation, currency, policy. Sets the tailwind or headwind (e.g., rate cuts favour rate-sensitives). 2. **Sector/industry** — pick sectors positioned to benefit; assess industry structure and cycle. 3. **Stock** — pick the best companies within the chosen sectors.

Bottom-up analysis (company first): start with the individual company's fundamentals — competitive position, financials, management, valuation — largely regardless of the macro call, on the belief that a great business at a fair price wins over time. Most research combines both.

Analysing the company — the toolkit: | Dimension | What to assess | |---|---| | **Revenue model & unit economics** | How it makes money; per-unit profitability | | **Competitive position / moat** | Brand, cost, switching costs, network, scale, IP (Porter's five forces) | | **Growth drivers** | Volume, price, mix, new products, markets | | **Margins & operating leverage** | Profitability trajectory and its drivers | | **Returns** | ROE, ROIC vs cost of capital (value creation) | | **Balance sheet & cash flow** | Leverage, working capital, FCF generation | | **Management & governance** | Track record, capital allocation, alignment | | **Valuation** | Is the price sensible vs intrinsic value? |

Qualitative + quantitative. The numbers (financials, ratios) show *what* happened; the qualitative work (moat, management, industry) explains *why* and whether it's sustainable. Great analysis fuses both.

Growth vs value styles. *Value* investors seek cheap stocks relative to fundamentals; *growth* investors pay up for superior growth. Both are fundamental — they weight the drivers differently.

Catalysts. Fundamental value only matters to a trade if something will make the market recognize it — new products, margin inflection, deleveraging, management change, sector re-rating. Identifying catalysts turns a valuation gap into an actionable idea.

Worked examples

Example 1 — top-down chain. Macro: rates are falling and consumption is recovering → favour consumer discretionary. Sector: within it, organized retail is gaining share from unorganized. Stock: pick the best-positioned retailer with a scale/cost moat and clean balance sheet. The macro call, sector selection, and stock pick compound.

Example 2 — bottom-up quality. An analyst ignores the noisy macro and focuses on a company with a durable brand moat, 25% ROIC (well above its ~11% WACC → strong value creation), consistent FCF, and disciplined capital allocation. Even through cycles, a business earning far above its cost of capital compounds value — a classic bottom-up buy.

Example 3 — good company, bad price. A wonderful franchise (high ROIC, wide moat) trades at 60× earnings, pricing in flawless growth for a decade. Fundamentally excellent, but the *price* already reflects it — the fundamental analyst concludes the stock, not the business, is the problem, and passes. Quality ≠ a buy at any price.

How it is tested in interviews

- **"What is fundamental analysis?"** — "Estimating a company's intrinsic value from its underlying economics — the macro, the industry, and the company's own drivers, quality and valuation."

- **"Top-down vs bottom-up?"** — "Top-down starts from the economy and sector and drills to the stock; bottom-up starts with the company's fundamentals. Most research combines both."
- **"How would you analyse a company's competitive position?"** — Porter's five forces / moat sources (brand, cost, switching costs, network, scale, IP) and returns vs cost of capital.
- **"How do you know a company creates value?"** — "ROIC above WACC — it earns more than the cost of the capital it uses."
- **"Why do you need a catalyst?"** — "A valuation gap only pays off if something makes the market recognize it — new products, a margin inflection, deleveraging, a re-rating."

Traps & common mistakes

- Doing only **one level** (company without industry, or macro without the company).
- Confusing a **great business** with a **great stock** — price matters.
- All numbers, no **qualitative** judgment (moat, management, sustainability).
- Ignoring **ROIC vs WACC** — growth without returns above cost of capital destroys value.
- No **catalyst** — a permanently "cheap" stock can stay cheap.

First-principles recap

- Fundamental analysis derives intrinsic value from economy, industry and company.
- **Top-down** (macro → sector → stock) and **bottom-up** (company first) — usually both.
- Fuse quantitative (financials, ROIC) with qualitative (moat, management).
- Value creation = **ROIC > WACC**; a great business isn't a buy at any price.
- Catalysts turn a valuation gap into an actionable idea.

Quick-reference

Level	Focus
Macro	Growth, rates, inflation, policy
Sector	Structure, cycle, competition
Company	Moat, drivers, margins, returns, management
Value creation	ROIC > WACC
Catalyst	What makes the market re-rate

Three-Statement Modeling for Research

The Problem / Why this matters

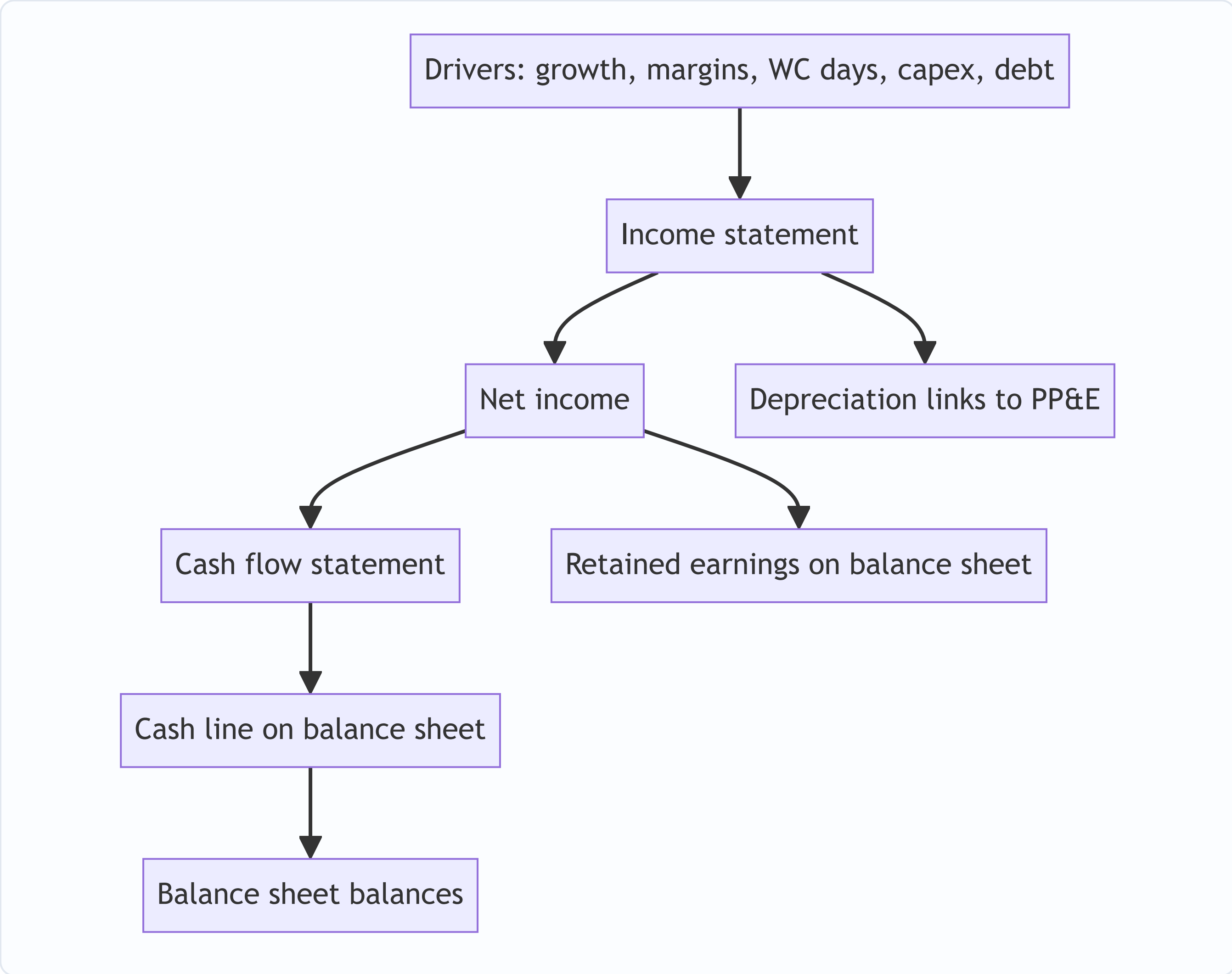
A research view needs numbers: a forecast of what the company will earn and how much cash it will generate. The **three-statement model** — a linked income statement, balance sheet and cash flow driven off a few assumptions — is the workhorse that produces those numbers and feeds valuation. Building and defending one is a core equity-research and IB skill, and "walk me through a 3-statement model" is a standard interview question.

Core Idea

A three-statement model projects the **income statement, balance sheet and cash flow statement**, fully linked so they stay internally consistent, from a set of **operating drivers** (revenue growth, margins, working-capital days, capex, financing). Change an assumption and the whole model — including free cash flow — updates coherently.

Why it works this way

The three statements are connected in reality (net income flows to retained earnings and to cash flow; depreciation links the P&L and the asset base; debt links financing and interest), so a credible forecast must respect those links. Driving everything off a small set of assumptions makes the model transparent, defensible, and easy to sensitize.



Full technical content

Build order (typical): 1. **Revenue** — from drivers (volume × price, or growth rate; segment build). 2. **Costs & margins** — COGS and opex as % of revenue or per-unit; get to EBITDA and EBIT. 3. **Supporting schedules** — depreciation (off PP&E + capex), working capital (off days: DSO, DIO, DPO), debt & interest, equity. 4. **Complete the income statement** — interest (from the debt schedule), taxes, net income. 5. **Build the cash flow** — start from net income, add back D&A, adjust for working-capital changes (from the WC schedule), subtract capex, add/subtract financing. 6. **Complete the balance sheet** — cash (from CF), PP&E (prior + capex − depreciation), working-capital items, debt, retained earnings (prior + NI − dividends). 7. **Check it balances** — assets = liabilities + equity every year; cash ties to the cash-flow statement.

The links that make it consistent: - Net income → retained earnings (BS) and top of cash flow (CF). - Depreciation → reduces PP&E (BS), non-cash add-back (CF), expense (IS). - Capex → increases PP&E (BS), investing outflow (CF). - Working-capital changes → BS balances and CF operating adjustment. - Debt → BS balance, interest on the IS, financing flows on the CF.

Circularity. Interest depends on debt, debt depends on cash, cash depends on interest — a circular reference. Handled with **iterative calculation** (Excel's circular switch) or a **circularity breaker** (average debt / copy-paste). A classic modeling interview question.

Best practices: separate **inputs** (assumptions) from **calculations**; drive off a few clear drivers; keep formulas consistent across columns; build **checks** (balance-sheet balances, cash ties to CF); make scenarios toggleable; label units. A clean, auditable model beats a clever one.

From model to valuation: the model outputs the **free cash flows** (FCFF/FCFE) that the DCF discounts and the **metrics** (EBITDA, EPS) that multiples apply to — so the model is the engine behind the valuation.

Worked examples

Example 1 — driver-based revenue. A retailer: stores 100 growing to 120, revenue/store ₹5 cr growing 4%/yr. Year-1 revenue = 100 × 5 = ₹500 cr; Year-2 = 110 × 5.2 ≈ ₹572 cr. Costs at 85% of revenue → EBITDA 15% margin. The forecast flows from store count and sales density, not a guessed growth number — defensible and sensitizable.

Example 2 — the balancing check. Net income ₹70 cr, dividends ₹20 cr → retained earnings up ₹50 cr. Cash flow shows net change in cash +₹30 cr. On the balance sheet, cash rises ₹30 cr and retained earnings rise ₹50 cr; other moves (PP&E, debt, working capital) must net so that assets = liabilities + equity. If it doesn't balance, there's a linking error — the check catches it.

Example 3 — the circularity. Adding a revolver whose interest depends on the average debt balance creates a circular reference (interest ↔ debt ↔ cash). Turning on iterative calculation lets Excel converge; alternatively, compute interest on opening (not average) debt to break the loop. Interviewers love this: "where's the circular reference in a 3-statement model?"

How it is tested in interviews

- **"Walk me through a three-statement model."** — Build revenue and costs → schedules (depreciation, working capital, debt) → complete the income statement → cash flow from net income → balance sheet → check it balances. Emphasize the *links*.
- **"How are the three statements linked?"** — Net income → retained earnings and CF; depreciation → PP&E, IS and CF; capex → PP&E and CF; debt → BS, interest and financing.
- **"Where is the circular reference?"** — "Interest ↔ debt ↔ cash. Solved with iterative calculation or by using opening-debt interest."
- **"What checks do you build?"** — "The balance sheet must balance every year, and cash on the balance sheet must equal the cash-flow statement's ending cash."

Traps & common mistakes

- Hard-coding numbers instead of driving off **assumptions**.
- Forgetting a **link** (e.g., depreciation not flowing to PP&E) so the model doesn't balance.
- Not handling **circularity** (or leaving it broken).
- No **checks** — errors go undetected.
- Over-complex models no one (including you) can audit.

First-principles recap

- A 3-statement model links IS, BS and CF, driven off a few operating assumptions.
- Build revenue → costs → schedules → complete IS → CF → BS → **check it balances**.
- Net income, depreciation, capex, working capital and debt are the key links.

- Handle **circularity** (interest ↔ debt ↔ cash) with iteration or a breaker.
- The model outputs the FCF and metrics that feed valuation.

Quick-reference

Link	From → To
Net income	IS → retained earnings (BS) & CF
Depreciation	IS expense, CF add-back, PP&E ↓
Capex	CF outflow, PP&E ↑
Working capital	CF adjustment, BS items
Debt	BS, interest (IS), financing (CF)
Check	BS balances; CF cash = BS cash

Applied Equity Valuation

The Problem / Why this matters

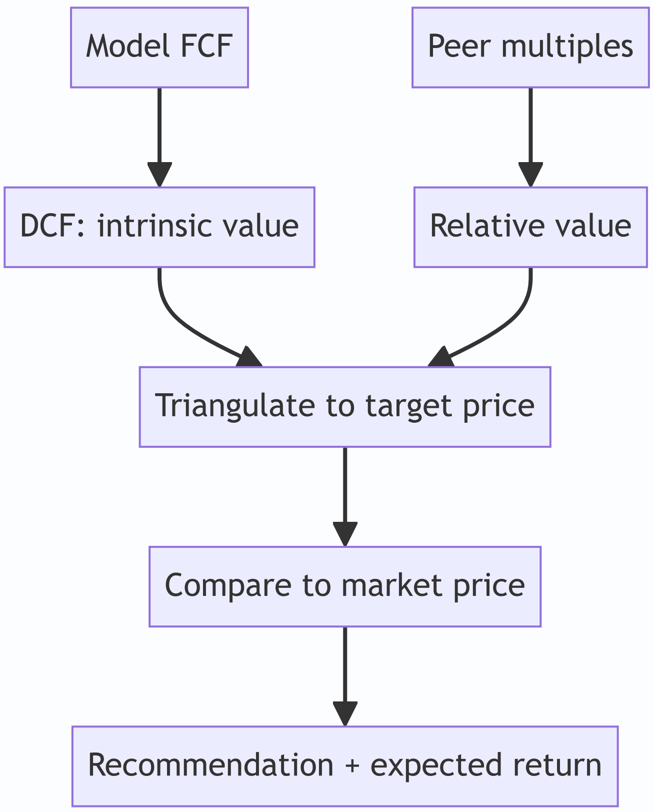
Research turns analysis into a **number** — an estimate of what a share is worth — so it can be compared to the market price. This chapter applies the valuation toolkit (covered deeply in the Valuation book) specifically to **equity research**: how analysts actually use DCF and multiples to set a target price and a recommendation. Interviewers test whether you can value a company end to end and defend the assumptions.

Core Idea

Applied equity valuation triangulates **intrinsic value** (DCF) and **relative value** (multiples vs peers) into a target price. The gap between that target and the market price — plus catalysts — drives the buy/sell/hold recommendation and the expected return.

Why it works this way

No single method is definitive: DCF is theoretically pure but assumption-sensitive; multiples reflect the market's current view but can be collectively wrong. Using both and cross-checking gives a value *range* you can defend, and forces you to understand *why* they differ.



Full technical content

DCF in research: - Project 5–10 years of **free cash flow** (FCFF) from the model. - Discount at **WACC**; add a **terminal value** (Gordon growth or exit multiple). - Sum to **enterprise value**; bridge to **equity value** (subtract net debt, minority interest; add associates); divide by diluted shares → **intrinsic price**. - Run **sensitivity** on WACC and terminal growth (the two swing factors) and present a range.

Relative valuation (multiples): - Pick the right multiple for the sector: **EV/EBITDA** (capital-structure neutral, most sectors), **P/E** (equity-level, mature/stable), **EV/Sales** (loss-making/high-growth), **P/B** (banks/financials), **PEG** (growth-adjusted). - Apply the peer multiple to the company's metric → implied value. - Adjust for differences in growth, margins, returns and risk vs peers.

Setting the target price. Weight the methods (e.g., 60% DCF, 40% comps), or use a football-field range and pick a point. The **12-month target price** implies an expected return vs the current price, which maps to the rating: | Expected return | Typical rating | |---|---| | > +15% | Buy / Overweight | | −10% to +15% | Hold / Neutral | | < −10% | Sell / Underweight |

Bank/financials valuation uses P/B and ROE (excess-return / residual-income models), and DDM — since FCFF/EV multiples don't work for financials.

High-growth/loss-makers rely on revenue multiples, unit economics, and a path-to-profitability DCF with explicit scenarios.

Sanity checks: does the implied terminal multiple make sense? Is terminal growth below GDP? Does the target imply a reasonable forward P/E? Reconcile DCF and comps — a large gap needs an explanation.

Worked examples

Example 1 — DCF to target. FCFF grows to ₹120 cr in year 5; WACC 11%, terminal growth 4%. Terminal value = 120 × 1.04 / (0.11 − 0.04) = ₹1,783 cr, discounted back; sum of PV of FCFs + PV of TV = enterprise value ₹1,400 cr. Less net debt ₹200 cr = equity value ₹1,200 cr; ÷ 100 mn shares = **₹120/share** intrinsic. Market price ₹100 → ~20% upside → **Buy**.

Example 2 — comps cross-check. Peers trade at 12× EV/EBITDA. The company's EBITDA is ₹130 cr → implied EV ₹1,560 cr; less net debt ₹200 cr = equity ₹1,360 cr ÷ 100 mn = **₹136/share**. The DCF said ₹120, comps say ₹136 — a defensible range of ₹120–136 vs a ₹100 price; the analyst sets a target around ₹128 and rates it Buy.

Example 3 — why DCF and comps differ. DCF (₹120) is below comps (₹136) because the analyst's forecast is more conservative than the growth the market is pricing into peers. Stating *why* they differ — and which you trust — is the analytical value-add. If you think the market's peer growth is too optimistic, lean on the DCF.

How it is tested in interviews

- **"How would you value this company?"** — "DCF for intrinsic value — project FCFF, discount at WACC, add terminal value, bridge EV to equity, per share — cross-checked with peer multiples (EV/EBITDA, P/E), then triangulate to a target price and compare to the market."
- **"Which multiple would you use for a bank?"** — "P/B with ROE, or a residual-income/DDM model — EV/EBITDA and FCFF don't work for financials."

- **"Your DCF says 120, comps say 136 — what do you do?"** — "Present the range, explain the difference (my forecast is more conservative than the growth peers price in), and lean on the method whose assumptions I trust more."
- **"How does the target price map to a rating?"** — "The implied 12-month return vs the current price: strong upside → Buy, roughly fair → Hold, downside → Sell."

Traps & common mistakes

- Relying on **one method** — always triangulate DCF and comps.
- Using the **wrong multiple** for the sector (EV/EBITDA on a bank).
- Terminal value assumptions that imply **absurd** growth or multiples.
- A target price with no **sensitivity** or range.
- Not explaining **why** DCF and comps differ.

First-principles recap

- Applied valuation triangulates **DCF (intrinsic)** and **multiples (relative)** into a target price.
- Bridge DCF enterprise value to equity value to a per-share number.
- Pick the sector-appropriate multiple; financials use P/B/ROE and DDM.
- The target's implied return vs the price sets the **rating**.
- Explain and reconcile any gap between methods.

Quick-reference

Method	Use
DCF (FCFF/WACC/TV)	Intrinsic value
EV/EBITDA	Most sectors, capital-neutral
P/E	Mature/stable, equity-level
P/B + ROE / DDM	Banks & financials
EV/Sales	Loss-making/high-growth
Target → rating	Implied return vs price

The Research Note & Investment Thesis

The Problem / Why this matters

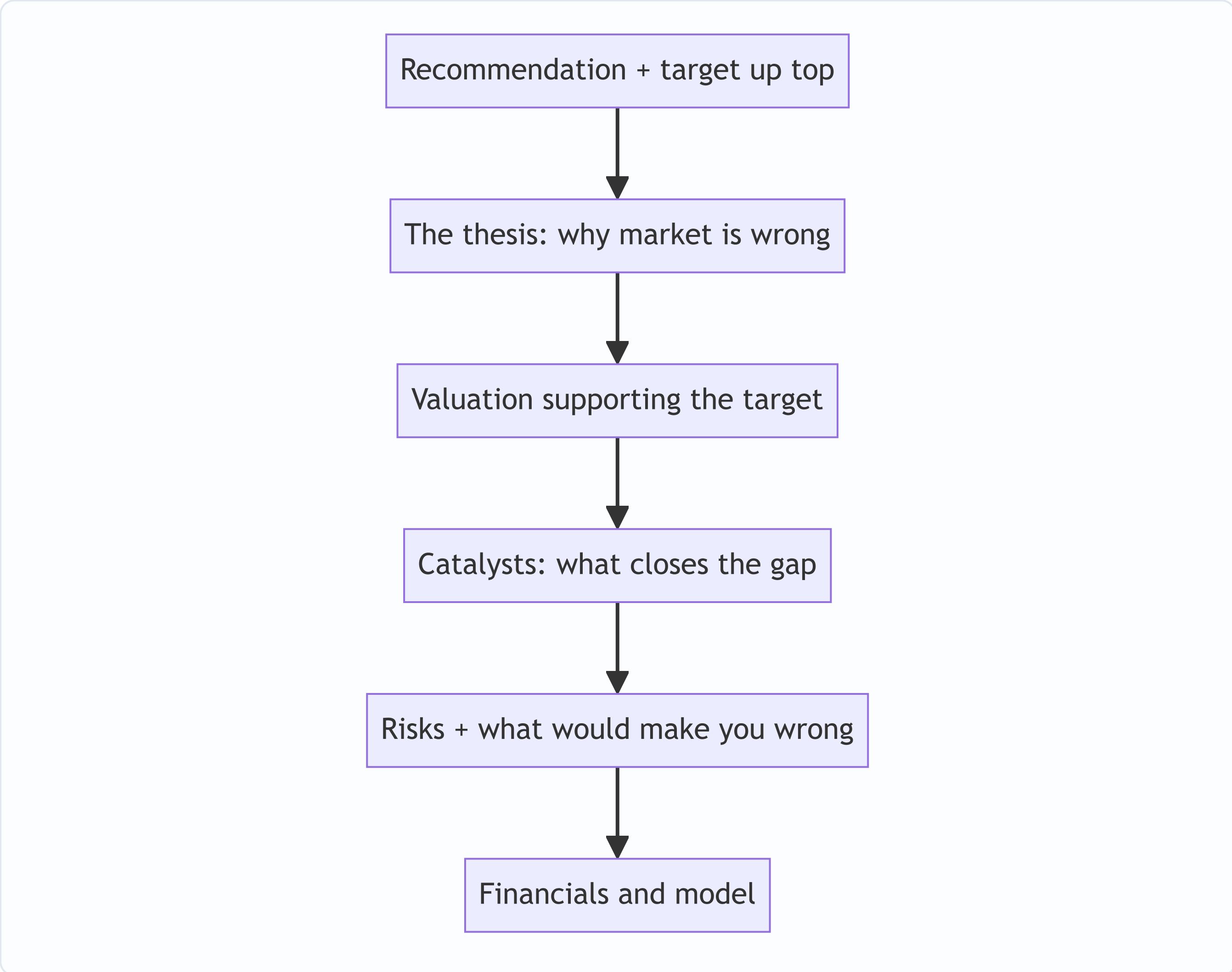
Analysis is worthless if it isn't communicated persuasively. The **research note** is the analyst's product — the written argument that a stock is mispriced, structured so a busy portfolio manager can grasp the thesis, valuation, catalysts and risks in minutes. Writing a tight note and articulating a crisp thesis is exactly what an equity-research interview tests (often via a written case or "pitch me a stock"). This chapter is the writing/communication counterpart to the analysis.

Core Idea

A research note communicates a **differentiated investment thesis** — why the market is wrong about this stock — supported by valuation, backed by catalysts, and honest about risks, ending in a clear **recommendation and target price**. Structure and clarity are as important as the analysis.

Why it works this way

PMs read dozens of notes; they act on the ones that are clear, differentiated, and defensible. A note that buries the thesis, agrees with consensus, or hides the risks gets ignored. So the note leads with the conclusion, states the *variant view*, quantifies the value, and pre-empts objections.



Full technical content

Types of note: initiation (full deep-dive launching coverage), **update/earnings note** (reaction to results or news), **thematic** (sector/idea). All share the core structure.

Anatomy of a research note: 1. **Recommendation & target** — rating (Buy/Hold/Sell), 12-month target price, and expected return — *at the top*. 2. **Investment thesis** — 3–5 crisp points on *why the market is wrong* and you're right (the variant view). 3. **Valuation** — the DCF and/or multiples supporting the target, with the key assumptions. 4. **Catalysts** — the specific events that will close the value gap and their timing. 5. **Risks** — the main downside risks and what would break the thesis. 6. **Business overview & financials** — the model, drivers, and key metrics.

The investment thesis — the heart. A great thesis is: - **Differentiated** — different from consensus (a variant perception); if you agree with the price, there's no trade. - **Specific & supported** — concrete drivers and numbers, not vague optimism. - **Falsifiable** — you can state exactly what would prove you wrong. - **Catalyst-driven** — a reason the market will come around.

The "variant view" test. Ask: *what do I believe that the market doesn't, and why am I right?* If you can't answer, you don't have a thesis — you have a description.

Writing craft: lead with the conclusion (BLUF — bottom line up front); be concise; use specific numbers; separate fact from opinion; make it skimmable (headers, bullets, a chart); and be intellectually honest about risks (it builds credibility).

Position on consensus. Note where you sit vs the Street (above/below consensus estimates and rating) — the value is in the *difference*, and being non-consensus and right is where alpha lives.

Worked examples

Example 1 — a crisp thesis (3 points). *Buy, target ₹1,200 (+20%).* Thesis: (1) The market underrates margin expansion from premiumization — we model 300 bps vs consensus 100 bps. (2) Rural distribution moat is widening, driving volume share gains consensus misses. (3) Valuation at 22× forward P/E is below the 5-year average despite better growth. Catalysts: next two quarters of margin beats. Risks: rural slowdown; raw-material inflation. *Differentiated, specific, catalyst-driven, falsifiable.*

Example 2 — description vs thesis. Weak note: "Company X is a good business with strong brands and steady growth; we rate it Buy." That's a *description* — it says nothing the market doesn't know and gives no reason the stock is mispriced. Strong note: "We are 15% above consensus on FY26 EPS because the Street hasn't modelled the new plant's margin uplift; at the current price that's not in the stock." The second has a variant view.

Example 3 — honest risk section building credibility. A Buy note explicitly says: "If Q1 volume growth comes in below 8% (vs our 15%), our margin thesis breaks and we would downgrade." Stating the falsification point makes the whole note more credible — the PM trusts an analyst who knows what would prove them wrong.

How it is tested in interviews

- **"Pitch me a stock."** — Lead with the recommendation and target, then 2–3 thesis points (your variant view), the valuation, catalysts, and risks — in that order, concisely.
- **"What makes a good research note?"** — "Conclusion up top, a differentiated and falsifiable thesis, valuation that supports the target, specific catalysts, and honest risks."
- **"What's an investment thesis?"** — "A variant view — what you believe that the market doesn't, and why you're right — with catalysts to close the gap."
- **"How is your view different from consensus?"** — Interviewers want to hear you position vs the Street; the value is in the difference.

Traps & common mistakes

- Writing a **description**, not a **thesis** (no variant view).
- Burying the **recommendation** instead of leading with it.
- Omitting **catalysts** (why does the gap close?) or **risks** (looks like cheerleading).
- Agreeing with **consensus/price** — no edge, no trade.
- Not stating what would **falsify** the thesis.

First-principles recap

- The research note is the analyst's product — clarity and structure matter as much as analysis.
- Lead with **recommendation and target**; then thesis, valuation, catalysts, risks.

- A thesis must be **differentiated, specific, falsifiable, and catalyst-driven**.
- Value lives in the **difference from consensus** — the variant view.
- Honest risk disclosure builds credibility.

Quick-reference

Section	Content
Top	Rating + target + expected return
Thesis	Variant view (why market is wrong)
Valuation	DCF/multiples supporting target
Catalysts	Events that close the gap + timing
Risks	Downside + falsification point
Thesis test	What do I believe that the market doesn't?

The Stock Pitch

The Problem / Why this matters

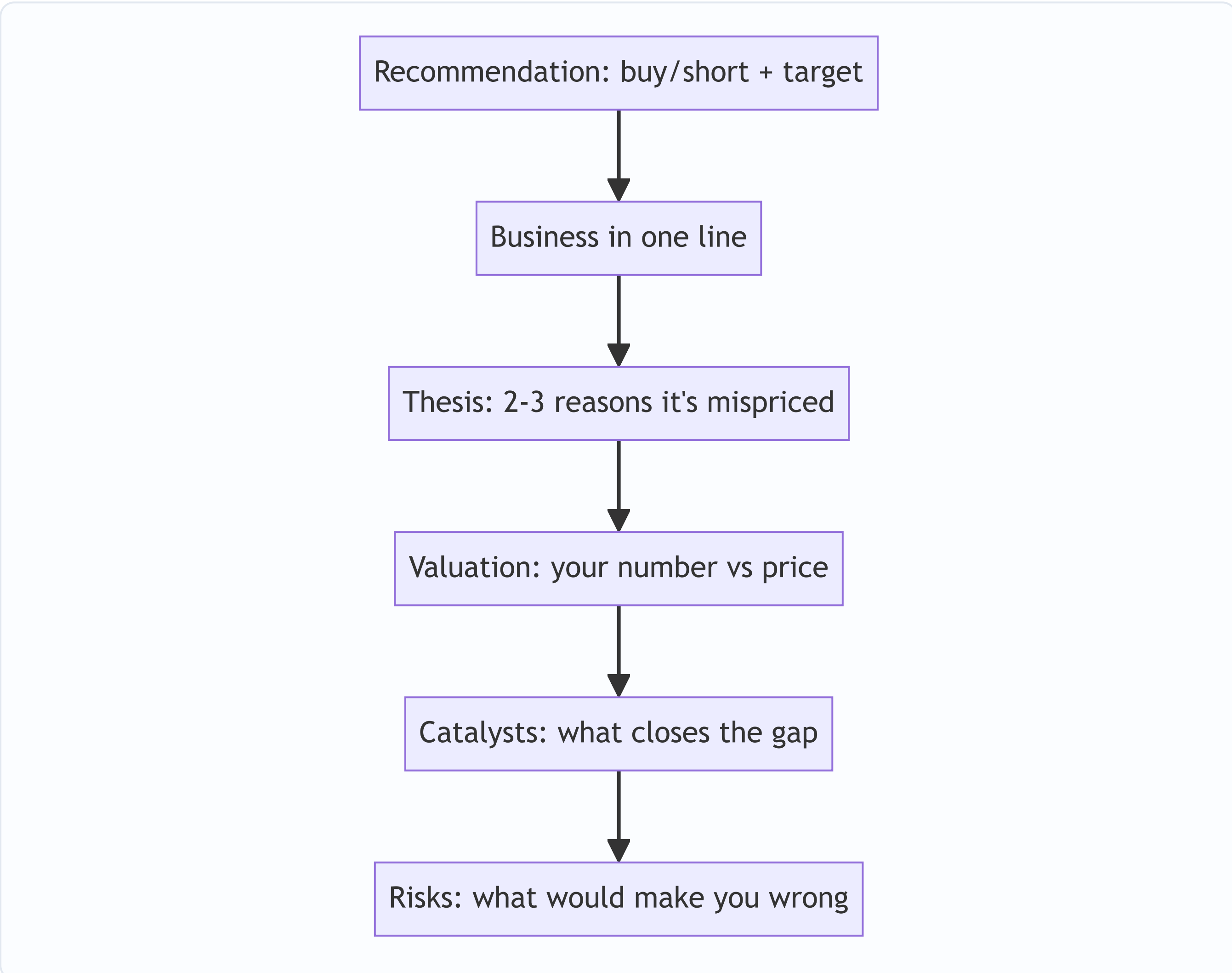
"Pitch me a stock" is the single most common question in equity research, buy-side, and markets interviews — and often the deciding one. It tests everything at once: whether you understand a business, can value it, have a differentiated view, know the catalysts and risks, and can communicate all of it in two minutes. A strong, structured pitch is a skill you can prepare and deliver on demand.

Core Idea

A stock pitch is a **concise, structured argument** for a specific action (buy or short) on a specific stock, delivered in a repeatable format: recommendation → business → thesis → valuation → catalysts → risks. It's the verbal version of the research note, compressed for a live conversation.

Why it works this way

The listener (interviewer or PM) wants to know quickly: *what's the trade, why is it mispriced, what's it worth, what makes it move, and what could go wrong.* Leading with the recommendation and following a clear skeleton signals that you think like an analyst and lets you cover everything without rambling.



Full technical content

The pitch skeleton (in order): 1. **Recommendation** — "I'd buy [X] with a 12-month target of ₹[Y], about [Z]% upside." (Or short.) Lead with it. 2. **Business** — one or two sentences: what the company does and how it makes money. 3. **Thesis** — 2–3 crisp reasons the market is mispricing it (your variant view). This is the core. 4. **Valuation** — your intrinsic value/target and how you got there (DCF and/or a multiple), vs the current price. 5. **Catalysts** — the specific events that will make the market recognize the value, with rough timing. 6. **Risks** — the main risks and what would prove you wrong (and ideally why the risk/reward is still favourable).

Delivery principles: - **Lead with the conclusion**; don't build up to it. - Be **specific** — real numbers (target, upside, key metric), not vague adjectives. - Have a **differentiated view** — say how you differ from consensus. - Keep it **~2 minutes**; then handle follow-up questions. - Show **balance** — acknowledging risks makes you credible.

Long vs short pitches. A **long** pitch argues undervaluation + positive catalysts. A **short** pitch argues overvaluation, deteriorating fundamentals, or a specific negative catalyst (accounting red flags, structural decline) — and must address the asymmetry (shorts have unlimited downside, borrow cost, and squeeze risk).

Preparation. Have **2–3 pitches ready** (ideally one long, one short), including at least one you genuinely follow. Know the numbers cold (valuation, key drivers) because the follow-ups probe them. Be ready for: "What's the market missing?", "What would make you wrong?", "What's it worth and why?", "What's the catalyst?"

Common follow-ups and how to handle them: | Follow-up | What they're testing | |---|---| | "Why is it mispriced?" | Your variant view / edge | | "What's your target and how?" | Valuation rigour | | "What's the catalyst?" | Actionability | | "What could go wrong?" | Intellectual honesty | | "Why hasn't the market seen this?" | Depth of edge |

Worked examples

Example 1 — a two-minute long pitch. "I'd buy [FMCG co], target ₹1,200, ~20% upside. It's a branded consumer-staples leader with a rural distribution moat. Thesis: the market underrates margin expansion from premiumization — I model 300 bps versus consensus 100 — and volume share gains from distribution the Street isn't crediting; at 22× forward it's below its historical average despite better growth. On a DCF and comps I get ₹1,200 versus ₹1,000 today. Catalysts: the next two quarters of margin beats. Risks: a rural slowdown or raw-material inflation — but at this valuation, I think the risk/reward is favourable." *Every element, two minutes.*

Example 2 — a short pitch. "I'd short [X], target 30% downside. It's a lender growing loans 40% a year while under-provisioning; my thesis is that credit costs are being deferred, not avoided — restated for realistic provisions, earnings are ~40% lower than reported. Valuation at 4× book assumes durable high ROE that I think is illusory. Catalyst: rising NPAs over the next two quarters and a likely provisioning reset. Risks: it can stay expensive and squeeze shorts, so I'd size it carefully and use a stop." *Addresses the short's asymmetry.*

Example 3 — handling "what would make you wrong?" For the FMCG long: "If the next two quarters show margin *contraction* instead of the expansion I expect — say raw-material inflation the company can't pass through — my core thesis breaks and I'd exit." A crisp falsification point signals a real analyst.

How it is tested in interviews

- **"Pitch me a stock."** — Deliver the skeleton: recommendation + target → business → 2–3 thesis points → valuation → catalysts → risks, in ~2 minutes.
- **"What's the market missing?"** — Your variant view — the single most important thing; have it sharp.
- **"What's it worth and how did you get there?"** — Give a target with a one-line valuation method; know the numbers.
- **"What would make you wrong?"** — A specific falsification point, delivered confidently.
- **"Long or short something you follow"** — Always have real pitches ready, not textbook examples.

Traps & common mistakes

- **Burying the recommendation** — lead with it.
- No **variant view** — just describing a "good company."
- **Vague** — no target, no numbers.
- Ignoring **risks** — sounds naive; acknowledging them builds credibility.
- Not knowing your own numbers when **probed**.
- For shorts, ignoring **borrow cost, squeeze, and unlimited downside**.

First-principles recap

- A pitch = recommendation → business → thesis → valuation → catalysts → risks, in ~2 minutes.
- **Lead with the conclusion**; be specific with numbers.
- The thesis is a **variant view** — what the market is missing.
- Name **catalysts** (actionability) and **risks/falsification** (credibility).
- Prepare 2–3 real pitches and know the numbers cold.

Quick-reference

Element	One line
Recommendation	Buy/short + target + upside
Business	What it does, how it earns
Thesis	2–3 reasons it's mispriced (variant view)
Valuation	Your number vs price + method
Catalysts	What closes the gap + when
Risks	Downside + what proves you wrong

Technical Analysis Essentials

The Problem / Why this matters

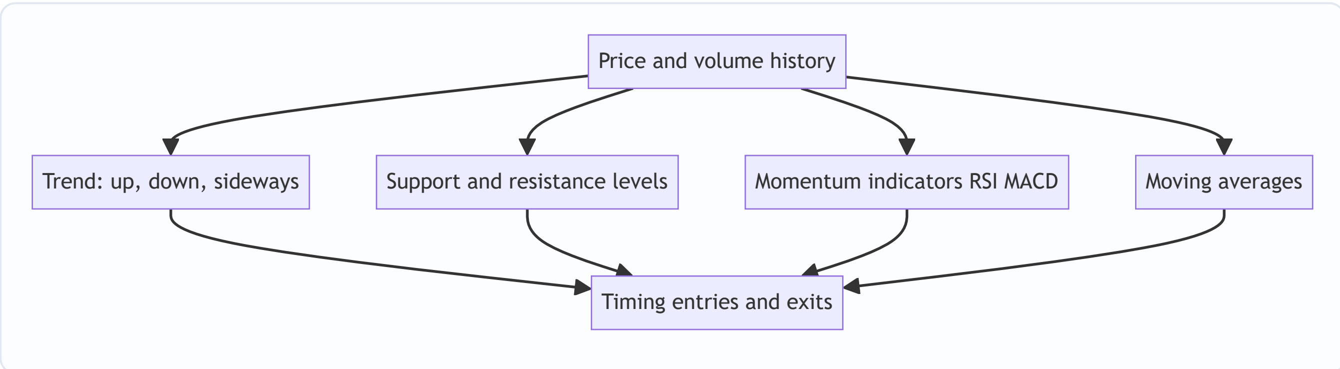
While fundamental analysis asks *what a stock is worth*, **technical analysis** asks *what the price is likely to do next*, from the price and volume history itself. It's the core toolkit for traders and technical research analysts, and it's used alongside fundamentals for timing entries and exits. Even fundamentally-driven roles expect you to understand trends, support/resistance, and the main indicators — and for trading/TRA roles it's central.

Core Idea

Technical analysis studies **price and volume patterns** to forecast future price movement, resting on three assumptions: the price discounts everything, prices move in trends, and history tends to repeat (because human behaviour does). Its tools identify trends, key price levels, momentum, and reversals.

Why it works this way

Price reflects the collective actions of all participants, so patterns in price can reveal the balance of supply and demand and the psychology (fear/greed) driving it. Trends persist because participants react and herd; levels matter because memory and orders cluster at them; patterns repeat because human behaviour under uncertainty is repetitive.



Full technical content

The three assumptions of technical analysis: 1. **The market discounts everything** — all known information is already in the price, so study the price. 2. **Prices move in trends** — once established, a trend is more likely to continue than reverse. 3. **History repeats** — patterns recur because market psychology is consistent over time.

Trend. The primary concept: an **uptrend** = higher highs and higher lows; a **downtrend** = lower highs and lower lows; **sideways/range** = no clear direction. "The trend is your friend" — trade with it until it clearly breaks. **Trendlines** connect the lows (uptrend) or highs (downtrend).

Support & resistance. Support = a price level where buying tends to emerge and halt declines; **resistance** = where selling tends to emerge and cap advances. A broken resistance often becomes new support (and vice versa). These levels are where traders place orders and stops.

Moving averages (MA). Smooth price to reveal the trend. Simple (SMA) or exponential (EMA). A price above its MA is bullish; below is bearish. Crossovers signal trend changes: a **golden cross** (50-day crosses above 200-day) is bullish; a **death cross** (50-day below 200-day) is bearish.

Momentum / oscillators: | Indicator | What it shows | |---|---| | **RSI** (Relative Strength Index) | Momentum, 0–100; > 70 overbought, < 30 oversold; divergences signal reversals | | **MACD** (Moving Average Convergence Divergence) | Trend + momentum via two EMAs and a signal line; crossovers and histogram | | Stochastic | Momentum vs recent range | | Bollinger Bands | Volatility bands around an MA |

Volume confirms moves — a breakout on high volume is more reliable than on low volume. **Chart patterns** (head-and-shoulders, double top/bottom, triangles, flags) signal continuation or reversal.

Technical vs fundamental — complementary. Fundamentals answer *what* to buy (value); technicals answer *when* (timing). Many analysts use fundamentals for the thesis and technicals for entry/exit and risk (stops).

Worked examples

Example 1 — trend and trendline. A stock makes higher highs and higher lows for months (uptrend); a trendline drawn under the rising lows holds. A trader buys on pullbacks to the trendline with a stop just below it, riding the trend until the line breaks — a break signalling the uptrend may be over.

Example 2 — support turned resistance. A stock repeatedly bounces off ₹100 (support). It finally breaks below to ₹90. On the next rally, ₹100 now acts as *resistance* — sellers who were trapped exit there. The old floor becomes the new ceiling, a classic level flip.

Example 3 — RSI divergence. Price makes a new high but RSI makes a *lower* high (bearish divergence) — momentum is weakening even as price rises, warning of a possible reversal. Combined with resistance, it's a signal to tighten stops or take profit.

How it is tested in interviews

- **"What is technical analysis and its assumptions?"** — "Studying price and volume to forecast prices, on three assumptions: the price discounts everything, prices move in trends, and history repeats."
- **"Technical vs fundamental analysis?"** — "Fundamentals estimate intrinsic value (what to buy); technicals study price/volume for direction and timing (when). They're complementary."
- **"What is support and resistance?"** — "Levels where buying (support) or selling (resistance) tends to emerge; broken resistance often becomes support."
- **"What do RSI and MACD tell you?"** — "RSI is a 0–100 momentum gauge (>70 overbought, <30 oversold); MACD combines trend and momentum via EMA crossovers. Divergences warn of reversals."
- **"What's a golden cross?"** — "The 50-day MA crossing above the 200-day — a bullish trend signal; the reverse (death cross) is bearish."

Traps & common mistakes

- Treating technicals as **certainty** — they're probabilities, not guarantees.
- Ignoring **volume** confirmation of breakouts.
- Using indicators in **isolation** rather than confluence (trend + level + momentum).
- Fighting the **trend** ("catching a falling knife").

- Forgetting technicals are best **combined** with fundamentals and risk management (stops).

First-principles recap

- Technical analysis forecasts price from price/volume history.
- Three assumptions: price discounts everything, trends persist, history repeats.
- Core tools: **trend**, **support/resistance**, **moving averages**, **RSI/MACD**, **volume**.
- Golden/death crosses and divergences signal trend/momentum shifts.
- Best used **with** fundamentals — value for *what*, technicals for *when*.

Quick-reference

Tool	Signal
Trend	Higher highs/lows = up; trade with it
Support/resistance	Buy/sell levels; break flips role
Moving average	Above = bullish; golden/death cross
RSI	>70 overbought, <30 oversold, divergence
MACD	EMA crossover, momentum
Volume	Confirms breakouts

Market Efficiency & Behavioral Finance

The Problem / Why this matters

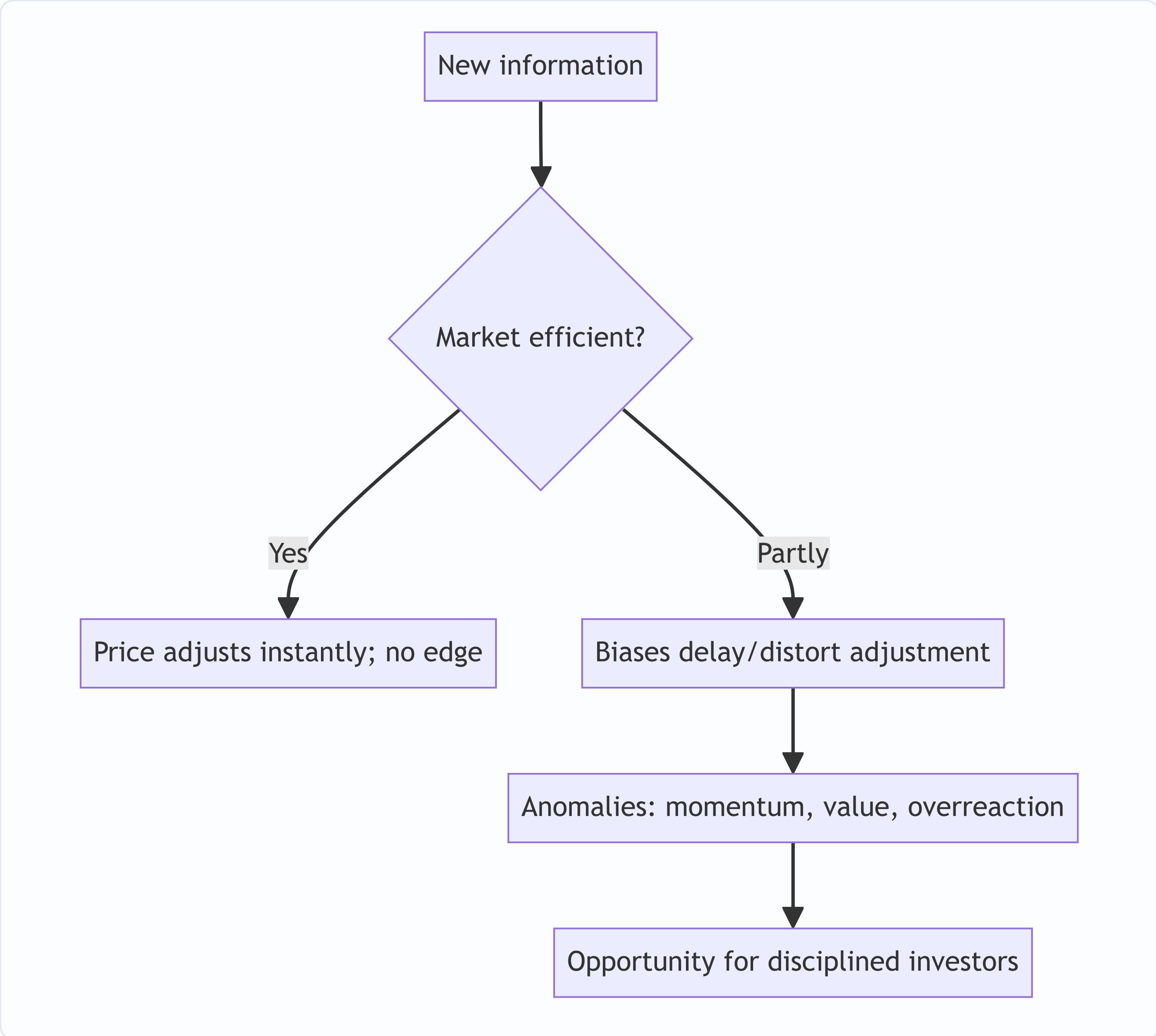
Can you beat the market? The answer shapes everything an investor does. The **Efficient Market Hypothesis (EMH)** says prices already reflect information, so consistently beating the market is nearly impossible — while **behavioral finance** documents the systematic biases that make markets *inefficient* enough to create opportunities. Every research and investing role sits in the tension between these two, and interviewers test whether you understand both.

Core Idea

The **EMH** holds that asset prices reflect available information, so you can't consistently earn risk-adjusted excess returns. **Behavioral finance** counters that investors are systematically irrational — subject to biases that push prices away from fundamentals — creating anomalies a disciplined investor can exploit. Reality lies between: markets are *mostly* efficient but not perfectly.

Why it works this way

If a piece of information predicted higher returns, rational investors would trade on it until the opportunity vanished — that's the efficiency argument. But real investors are human: they overreact, herd, anchor and fear losses, and those biases are *systematic* (not random), so they don't fully cancel out — leaving persistent, if hard-to-capture, mispricings.



Full technical content

The three forms of EMH: | Form | Prices reflect | Implication | |---|---|---| | **Weak** | All past prices/volume | Technical analysis shouldn't work | | **Semi-strong** | All public information | Fundamental analysis on public data shouldn't give an edge | | **Strong** | Even private/inside information | No one can beat the market, even insiders |

Evidence: markets are *broadly* semi-strong efficient (prices react fast to news), but **anomalies** persist that challenge it.

Anomalies (evidence against strict EMH): - **Value effect** — cheap (low P/B, P/E) stocks historically outperform. - **Momentum** — recent winners keep winning over 3–12 months. - **Small-cap effect**, **low-volatility anomaly**, **post-earnings-announcement drift**, calendar effects. These underpin factor investing (Fama-French).

Behavioral biases (why markets misprice): | Bias | Effect | |---|---| | **Overconfidence** | Excess trading, underestimating risk | | **Anchoring** | Fixating on a reference price | | **Loss aversion** | Losses hurt ~2× gains; holding losers, selling winners (disposition effect) | | **Herding** | Following the crowd → bubbles and crashes | | **Recency bias** | Overweighting recent events | | **Confirmation bias** | Seeking evidence that fits your view | |

Mental accounting | Treating money differently by source/label | | **Overreaction / underreaction** | To news → reversals and drift |

Bubbles and crashes are the macro expression of these biases — herding and overconfidence inflate prices far above value, then fear collapses them.

Implications for investors: - If markets were perfectly efficient → **index passively** (no edge to be had). - Because they're not perfectly efficient → disciplined, unemotional investors can exploit others' biases (the behavioral edge), but it's hard and requires patience. - Know your *own* biases — the first job is not to be the sucker.

Worked examples

Example 1 — semi-strong efficiency in action. A company reports earnings 20% above expectations at 9:00 am; by 9:01 the stock has jumped to reflect it. A retail investor reading the news at noon can't profit from the surprise — it's already priced. That's semi-strong efficiency: public information is incorporated almost instantly.

Example 2 — loss aversion / disposition effect. An investor holds a stock down 30%, refusing to sell "until it comes back," while quickly booking a 10% gain elsewhere. Loss aversion makes them ride losers and cut winners — exactly backwards. Recognizing this bias (and using rules/stops) is the behavioral edge.

Example 3 — herding bubble. During a mania, prices detach from any fundamental value as investors buy because others are buying (herding) and extrapolate recent gains (recency). Fundamentals say "expensive," but the crowd pushes higher — until sentiment turns and it crashes. Bubbles are behavioral finance writ large.

How it is tested in interviews

- **"What is the Efficient Market Hypothesis?"** — "Prices reflect available information, so you can't consistently earn risk-adjusted excess returns. Three forms: weak (past prices), semi-strong (public info), strong (private info)."
- **"If markets are semi-strong efficient, what stops working?"** — "Trading on public information/fundamental analysis of public data — it's already in the price. And technical analysis fails under even the weak form."
- **"Do you believe markets are efficient?"** — "Broadly semi-strong efficient — news is priced fast — but not perfectly; behavioral biases and documented anomalies (value, momentum) leave exploitable inefficiencies."
- **"Name a behavioral bias and its effect."** — Loss aversion → holding losers too long (disposition effect); herding → bubbles; anchoring → fixating on a reference price.
- **"If markets were perfectly efficient, how would you invest?"** — "Passively, via low-cost index funds — there'd be no edge to justify active fees."

Traps & common mistakes

- Treating EMH as **all-or-nothing** — it's a spectrum; markets are mostly-but-not-perfectly efficient.
- Confusing the **three forms** (weak/semi-strong/strong).
- Assuming biases **cancel out** — they're systematic, not random.

- Believing anomalies are **free money** — they're real but hard and risky to capture.
- Ignoring your **own** biases.

First-principles recap

- EMH: prices reflect information → hard to beat the market; forms are weak/semi-strong/strong.
- Behavioral finance: systematic biases push prices from value, creating anomalies.
- Reality: mostly efficient, not perfectly — news is priced fast, but inefficiencies persist.
- Anomalies (value, momentum) underpin factor investing.
- The behavioral edge is discipline; the first task is not to be biased yourself.

Quick-reference

Concept	Note
Weak EMH	Past prices priced → TA fails
Semi-strong	Public info priced → FA edge hard
Strong	Even private info priced
Anomalies	Value, momentum, small-cap, PEAD
Key biases	Loss aversion, herding, anchoring, overconfidence
If perfectly efficient	Index passively

Corporate Actions

The Problem / Why this matters

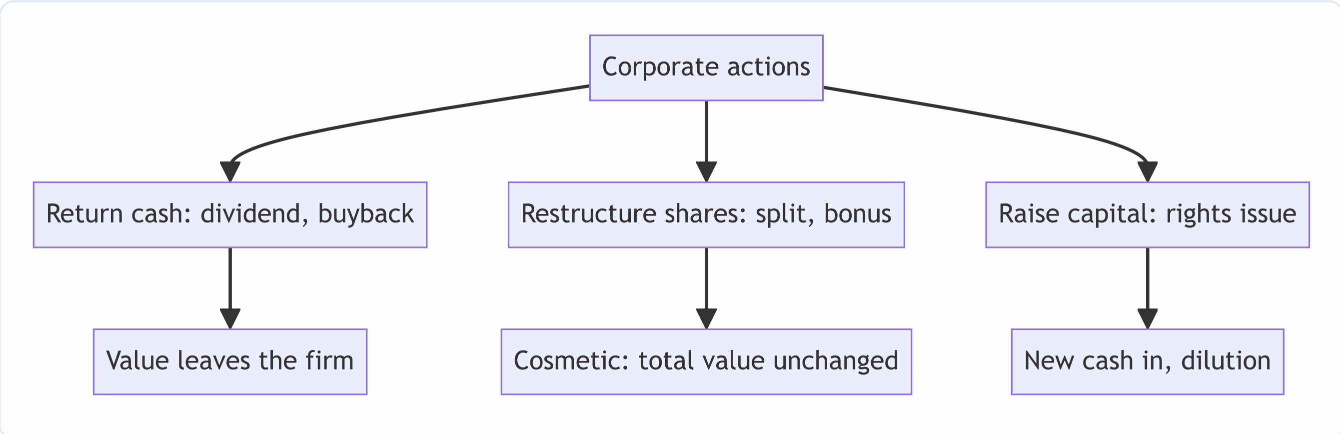
Companies regularly do things that change their shares — pay dividends, split the stock, issue bonus or rights shares, buy back stock. These **corporate actions** change share counts, prices, and per-share metrics, and an analyst must adjust for them correctly (in models, historical price series, and valuation) — and know which ones create value and which merely rearrange it. Interviewers test whether you know that a split or bonus doesn't change value, and how a buyback flows through.

Core Idea

A **corporate action** is an event initiated by a company that affects its securities and shareholders. Some return cash (dividends, buybacks); some change the share structure without changing total value (splits, bonus issues); some raise capital (rights issues). Knowing the mechanics and the value impact of each is essential.

Why it works this way

Per-share numbers depend on the share count, so any action that changes shares changes per-share metrics — but often not total value. A split cuts the price and multiplies shares proportionally (same market cap); a dividend moves cash from the company to shareholders (value leaves the firm). Distinguishing *value-changing* from *cosmetic* actions is the key skill.



Full technical content

Dividends. Cash paid per share from profits/reserves. Reduces cash and retained earnings. Key dates: **declaration, ex-dividend** (buy before ex-date to receive it), **record**, **payment**. The price typically drops by ~the dividend on the ex-date. **Dividend yield** = dividend/price.

Stock split. Divides each share into more shares at a proportionally lower price — e.g., a **1:1 split** (₹100 → two shares of ₹50). Market cap, and your total value, **unchanged**; only the share count and price change. Purpose: improve affordability and liquidity. (A **reverse split** consolidates shares into fewer, higher-priced ones.)

Bonus / scrip issue. Free additional shares to existing holders from reserves (e.g., **1:1 bonus** = one free share per share held). Like a split, it increases shares and cuts the price proportionally — **no change in total value** (reserves just move to share capital). Purely accounting/signaling.

Rights issue. Existing shareholders get the *right* to buy new shares, usually at a **discount**, in proportion to holdings (e.g., 1 new share per 4 held at a discount). Raises capital; those who don't subscribe are **diluted** (though the right itself has value and can be sold). The price adjusts to a **theoretical ex-rights price (TERP)** blending old and new.

Buyback (share repurchase). The company buys its own shares (tender or open-market) and cancels them. Reduces cash and equity, shrinks share count → **boosts EPS**. Returns cash like a dividend but more tax-efficiently/flexibly; signals shares are cheap. Doesn't create value by itself (depends on price paid vs value and alternative uses of cash).

Action	Shares	Price/share	Total value	Cash impact
Dividend	Unchanged	Falls ~dividend	Falls (cash out)	Cash out
Split (1:1)	×2	÷2	Unchanged	None
Bonus (1:1)	×2	÷2	Unchanged	None
Rights	Up	TERP (blend)	Up by cash raised	Cash in
Buyback	Down	~Unchanged	Falls (cash out)	Cash out

Analyst adjustments. Adjust **historical prices** for splits/bonus (so charts aren't distorted), adjust **share counts** in models, and use **diluted** shares. Never compare a pre-split price to a post-split price without adjusting.

Worked examples

Example 1 — a split creates no value. A stock at ₹1,000 does a 1:5 split → 5 shares at ₹200 each. An investor who held 10 shares (₹10,000) now holds 50 shares (₹10,000). Nothing changed except affordability. *A 2-for-1 or 5-for-1 split does not make you richer.*

Example 2 — buyback and EPS. A firm has net income ₹100 cr and 100 mn shares → EPS ₹10. It buys back 10 mn shares (cancelled) → 90 mn shares → EPS ₹100 cr / 90 mn = **₹11.1**. EPS rose ~11% with no change in profit — purely from a smaller share count. (Whether it *created value* depends on whether the shares were bought below intrinsic value.)

Example 3 — rights issue TERP. A stock at ₹200; a 1:4 rights issue at ₹100. For every 4 old shares (₹800) you buy 1 new at ₹100 → 5 shares for ₹900 → TERP = ₹180. The price "falls" from ₹200 to ₹180, but that's the dilution of the discounted new shares, not a loss for subscribers.

How it is tested in interviews

- **"Does a 2-for-1 stock split create value?"** — "No — twice the shares at half the price; total value and market cap are unchanged. It improves affordability and liquidity."
- **"How does a buyback affect EPS and the statements?"** — "Cash and equity fall; share count shrinks, so EPS rises. It returns cash like a dividend but more flexibly/tax-efficiently. It only creates value if shares are bought below intrinsic value."

- **"Split vs bonus issue?"** — "Both increase shares and cut the price proportionally with no value change; a split changes the face value, a bonus capitalizes reserves into share capital."
- **"What is a rights issue and TERP?"** — "Existing holders can buy new shares at a discount; the price settles at a theoretical ex-rights price blending old and new. Non-subscribers are diluted."
- **"Why adjust historical prices for a split?"** — "So the chart and returns aren't distorted by the mechanical price change."

Traps & common mistakes

- Thinking a **split/bonus** makes shareholders richer — it's cosmetic.
- Confusing a **buyback's EPS boost** with value creation (depends on price paid).
- Forgetting **rights issues dilute** non-subscribers.
- Comparing pre- and post-split prices **without adjusting**.
- Missing the **ex-date** logic for dividends/entitlements.

First-principles recap

- Corporate actions change shares, price, and per-share metrics — often not total value.
- **Splits and bonus issues are cosmetic** (value unchanged).
- **Dividends and buybacks** return cash (value leaves the firm); buybacks lift EPS.
- **Rights issues** raise capital and dilute non-subscribers (price → TERP).
- Always adjust historical prices and share counts for splits/bonus.

Quick-reference

Action	Value effect
Dividend	Cash out; price drops ex-date
Split / bonus	Cosmetic; total value unchanged
Buyback	Cash out; EPS up; value if bought cheap
Rights issue	Capital in; dilutes non-subscribers; TERP
Adjust for	Splits/bonus in prices & share counts

Capital Raising: Follow-ons, Rights, QIPs & Block Deals

The Problem / Why this matters

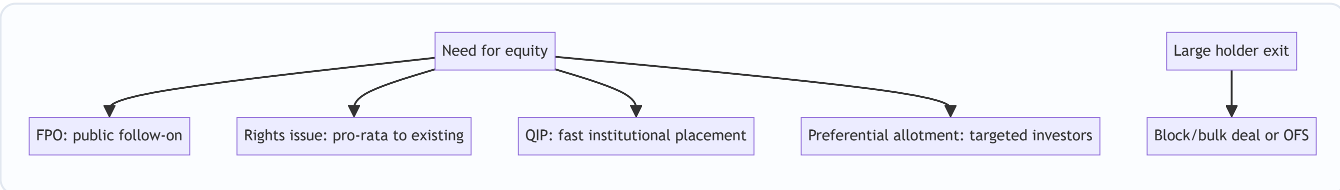
An IPO is a company's first equity raise, but listed companies raise equity again and again — to fund growth, cut debt, or let large holders exit. **Equity capital markets (ECM)** covers all these follow-on ways to issue or place equity: FPOs, rights issues, QIPs, preferential allotments, and block/bulk deals. Knowing each mechanism, when it's used, and its effect on existing shareholders is core to ECM, IB, and research roles.

Core Idea

Beyond the IPO, listed firms raise or place equity through several routes, differing by **who can buy, how fast, and at what dilution**: broad public offers (FPO), pro-rata offers to existing holders (rights), fast institutional placements (QIP), targeted placements (preferential), and secondary sales of existing shares (blocks/OFS). Each balances speed, price, investor base, and dilution.

Why it works this way

Different situations call for different tools: a company wanting a quick institutional raise uses a QIP (fast, no lengthy prospectus); one wanting to treat all shareholders fairly uses a rights issue; a large holder wanting to exit uses a block deal or OFS. The menu exists so issuers can optimize for speed, price, and the desired investor base.



Full technical content

Primary raises (new shares → company gets cash, dilution): | Route | Who buys | Speed | Notes | |---|---|---|---| | **FPO** | Public | Slow (prospectus) | A full follow-on public offer | | **Rights issue** | Existing holders, pro-rata | Moderate | Usually at a discount; non-subscribers diluted; price → TERP | | **QIP** | Qualified institutional buyers | **Fast** | India-specific; no lengthy prospectus; institutions only | | **Preferential allotment** | Selected investors | Fast | Targeted (e.g., a strategic/PE investor); priced per SEBI formula |

Secondary sales (existing shares → selling holder gets cash, no new capital, no dilution): | Route | What it is | |---|---| | **OFS (Offer for Sale)** | Promoters/large holders sell via an exchange mechanism | | **Block deal** | A large negotiated trade (min size) in a special window at a pre-agreed price | | **Bulk deal** | Large trades executed in the normal market (disclosed) |

Key distinctions: - **Fresh issue vs secondary sale** — fresh issue raises capital for the company and dilutes; secondary sale just transfers existing shares (holder gets cash, no dilution). - **Dilution** — new shares reduce existing holders' ownership % and EPS unless the capital earns above the cost of equity. - **Pricing** — placements are usually at a small discount to market to attract buyers; rights at a larger discount; SEBI sets floor-price formulas for QIPs/preferential. - **Signaling** — equity issuance can signal management thinks shares are fairly/over-valued (pecking-order); a raise to cut debt vs to fund high-return growth is read very differently.

ECM's job. Investment banks (ECM desks) advise on the route, timing, size, and pricing, and place the shares with investors — balancing the issuer's proceeds against a successful, well-received deal.

Worked examples

Example 1 — QIP for speed. A company needs ₹1,500 cr quickly to fund expansion. Rather than a months-long FPO, it does a **QIP**: places new shares with institutions at a ~3% discount to the market over a couple of days. Fast, institutional, and dilutive — the company gets the cash without a full prospectus process.

Example 2 — rights issue to deleverage. A leveraged firm does a **1:2 rights issue** at a 30% discount to cut debt. Existing holders can maintain their stake by subscribing; those who don't are diluted but can sell their rights. The stock re-rates to TERP; the balance sheet strengthens.

Example 3 — block deal exit (no dilution). A PE fund holding 8% wants to exit. It sells the entire stake in a single **block deal** to institutions at a small discount in the block window. The *fund* receives the cash; the company issues no new shares, so there's **no dilution** — just a change of ownership. (Contrast with a QIP, which is new shares and dilutive.)

How it is tested in interviews

- **"Ways a listed company can raise equity?"** — FPO, rights issue, QIP, preferential allotment (primary/dilutive); plus OFS/block deals for existing holders to sell (secondary, non-dilutive).
- **"What is a QIP and why use it?"** — "A Qualified Institutional Placement — a fast raise from institutions without a full prospectus; used when a company wants speed and an institutional base."
- **"Rights issue vs QIP?"** — "Rights is pro-rata to all existing holders (fair, at a discount, non-subscribers diluted); QIP is a quick placement to institutions only."
- **"Block deal vs QIP — is it dilutive?"** — "A block deal is a secondary sale of existing shares — the seller gets the cash, no dilution. A QIP is new shares — the company gets the cash, dilutive."
- **"Why might a raise move the stock down?"** — "Dilution, plus the signal that management may think shares are fully valued; the use of proceeds (debt reduction vs high-return growth) matters."

Traps & common mistakes

- Confusing **primary** (new shares, company cash, dilution) with **secondary** (existing shares, holder cash, no dilution).
- Thinking a **block deal/OFS** raises capital for the company — it doesn't.
- Forgetting **rights issues dilute** non-subscribers and reset the price to TERP.
- Ignoring the **signaling** and use-of-proceeds read of a raise.

First-principles recap

- Listed firms raise equity via FPO, rights, QIP, preferential (primary, dilutive) and sell via OFS/blocks (secondary, non-dilutive).

- **QIP** = fast institutional raise; **rights** = pro-rata to existing holders at a discount.
- Fresh issue → company cash + dilution; secondary sale → holder cash, no dilution.
- Placements price at a discount; SEBI sets floors.
- Read every raise for **dilution, signaling, and use of proceeds**.

Quick-reference

Route	Type	Cash to	Dilution
FPO	Primary	Company	Yes
Rights	Primary, pro-rata	Company	Yes (non-subscribers)
QIP	Primary, institutions	Company	Yes
Preferential	Primary, targeted	Company	Yes
OFS / block	Secondary	Selling holder	No

Sell-Side vs Buy-Side

The Problem / Why this matters

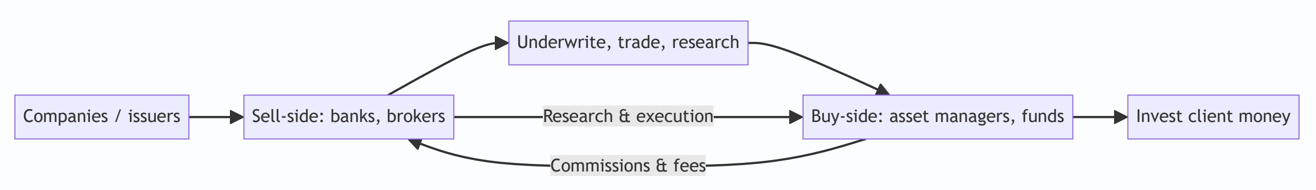
The finance industry splits into two worlds — the **sell-side** (banks and brokers who create, sell, and research products) and the **buy-side** (asset managers who invest client money). Every markets role sits on one side, they interact constantly, and interviewers frequently ask which you want and why. Understanding the distinction — who does what, how each makes money, and how research flows between them — is basic industry literacy.

Core Idea

The **sell-side** creates and sells financial products and services (underwriting, trading, research) to clients and earns fees/commissions. The **buy-side** buys and holds investments on behalf of clients (funds) and earns management/performance fees on assets. Sell-side research is *published* to win business; buy-side research is *private* and drives the fund's own decisions.

Why it works this way

The two exist because issuers and investors need different services. The sell-side intermediates — helping companies raise capital and helping investors execute and understand — monetized through transactions. The buy-side is the end investor of capital — monetized through investment performance on assets under management. Research flows from sell-side (marketing/service) to buy-side (decision input).



Full technical content

Sell-side (investment banks, brokerages): | Function | What it does | |---|---| | **Investment banking (ECM/DCM/M&A)** | Raise capital and advise on deals for companies | | **Sales & trading** | Execute trades, make markets, provide liquidity | | **Sell-side research** | Publish analysis and ratings on stocks/sectors to clients | Earns: **underwriting fees, advisory fees, trading commissions, spreads.** Sell-side research is *distributed* to institutional clients to win their trading/banking business — it's a service, not a profit centre itself.

Buy-side (asset managers): | Type | Description | |---|---| | **Mutual funds / AMCs** | Pooled long-only funds, benchmark-relative | | **Hedge funds** | Flexible, absolute-return, can short/leverage | | **Pension & insurance funds** | Long-term, liability-driven | | **PE / VC** | Private investing | | **PMS / AIFs** | Portfolio management for HNIs | Earns: **management fees (% of AUM)** and often **performance fees** (e.g., 2-and-20 for hedge funds). Buy-side research is *internal* and confidential — it directly drives what the fund buys.

How they interact: - Sell-side provides **execution** (trading), **capital-raising access** (IPO allocations), and **research** to the buy-side. - Buy-side pays via **commissions** and consumes sell-side research as one input (increasingly unbundled post-MiFID II). - The buy-side ultimately makes the investment decision; the sell-side facilitates.

Research differences: | | Sell-side research | Buy-side research | |---|---|---| | Audience | Published to many clients | Internal to the fund | | Purpose | Win business/commissions | Make the fund's own decisions | | Coverage | Broad, many names | Focused on holdings/ideas | | Incentive | Volume, access, relationships | Being *right* (performance) |

Career framing. Sell-side research/banking: broad coverage, client-facing, deal/idea flow. Buy-side: fewer names, deeper conviction, direct accountability for returns. Many move sell-side → buy-side over a career.

Worked examples

Example 1 — a research idea's journey. A sell-side analyst publishes a Buy on Stock X with a note. It's distributed to hundreds of institutional clients. A buy-side PM reads it, does her *own* work, disagrees on one assumption, but likes the idea — and buys through the same bank's trading desk (paying commission). The sell-side earned the commission and strengthened the relationship; the buy-side made the actual decision.

Example 2 — how each makes money. A brokerage (sell-side) earns ₹5 cr in commissions from a fund's trading this year. The fund (buy-side) manages ₹5,000 cr and earns a 1% management fee = ₹50 cr, plus performance fees if it beats its benchmark. Different economics: transaction fees vs fees on assets and performance.

Example 3 — incentive difference. A sell-side analyst is rewarded for coverage, client access, and generating trading interest; a buy-side analyst is rewarded purely for whether the calls make money. This is why buy-side research is more focused and conviction-driven — being *right* is the only thing that pays.

How it is tested in interviews

- **"What's the difference between sell-side and buy-side?"** — "Sell-side (banks/brokers) creates and sells products — underwriting, trading, research — for fees and commissions. Buy-side (asset managers) invests client money for management and performance fees. Sell-side research is published to win business; buy-side research is private and drives the fund's decisions."
- **"How does each make money?"** — Sell-side: underwriting/advisory fees, trading commissions/spreads. Buy-side: % of AUM management fees plus performance fees.
- **"Do you want sell-side or buy-side, and why?"** — Have a genuine answer: sell-side for broad coverage and client/deal exposure; buy-side for deep conviction and direct accountability for returns.
- **"How do sell-side and buy-side research differ?"** — Audience (published vs internal), purpose (win business vs make money), and incentive (volume/access vs being right).

Traps & common mistakes

- Mixing up which side **invests** (buy-side) vs **intermediates** (sell-side).
- Thinking sell-side research is the sell-side's **profit centre** — it's a service to win trading/banking.
- Not having a **genuine preference** with reasons when asked.
- Forgetting the buy-side does its **own** work and makes the final call.

First-principles recap

- **Sell-side** intermediates (underwrite, trade, research) for fees/commissions.
- **Buy-side** invests client money for fees on assets and performance.
- Research flows sell-side → buy-side; the buy-side decides.
- Incentives differ: sell-side rewards access/volume; buy-side rewards being right.
- Careers often move sell-side → buy-side.

Quick-reference

	Sell-side	Buy-side
Role	Create/sell/execute/research	Invest client money
Examples	IBs, brokerages	Mutual/hedge/pension funds, PMS
Revenue	Fees, commissions, spreads	% AUM + performance fees
Research	Published, wins business	Internal, drives decisions
Reward	Access/volume	Performance (being right)

Portfolio Construction & Risk

The Problem / Why this matters

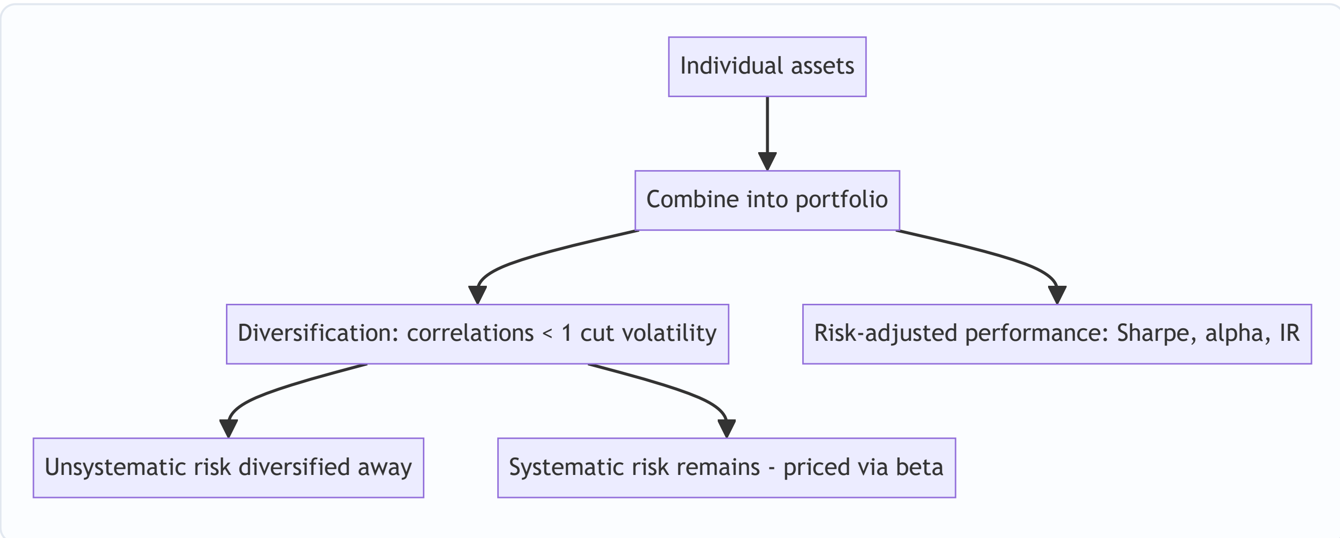
Owning a single great stock is risky; investing is really about building a **portfolio** — combining positions so the whole is better (higher return per unit of risk) than the parts. How you size positions, diversify, benchmark, and measure risk-adjusted performance is the core of asset management and any investing role. Interviewers test diversification, systematic vs unsystematic risk, beta, and the performance ratios.

Core Idea

Portfolio construction combines assets to maximize expected return for a given level of risk, exploiting **diversification** (imperfectly correlated assets reduce portfolio volatility). Risk splits into **systematic** (market, undiversifiable) and **unsystematic** (specific, diversifiable), and performance is judged on a **risk-adjusted** basis (Sharpe, alpha, information ratio), not raw return.

Why it works this way

Because asset returns aren't perfectly correlated, combining them means their ups and downs partly offset, lowering portfolio volatility without proportionally lowering return — the "only free lunch in finance." But the shared exposure to the economy (systematic risk) can't be diversified away, so it's what investors are ultimately compensated for (via beta and the risk premium).



Full technical content

Risk decomposition: - **Unsystematic (specific/idiosyncratic) risk** — company/sector-specific; **diversifiable** by holding many uncorrelated names. - **Systematic (market) risk** — common exposure to the economy; **not diversifiable**; measured by **beta**; it's what earns the risk premium.

Diversification. Combining assets with correlation < 1 reduces portfolio standard deviation for a given return. The benefit is largest when correlations are low/negative; most idiosyncratic risk is removed with ~20–30 well-chosen names, leaving mainly systematic risk.

Modern Portfolio Theory (Markowitz). For a given return, there's a minimum-variance portfolio; the set of best risk-return combinations is the **efficient frontier**. Adding a risk-free asset gives the **Capital Market Line**; the best risky portfolio is the market portfolio (→ CAPM).

Position sizing & construction approaches: - **Concentration vs diversification** — high-conviction concentrated books (fewer, larger positions) vs broadly diversified books; a trade-off between idiosyncratic risk and edge. - **Active vs passive** — active tries to beat a benchmark (higher fees, tracking error); passive replicates it cheaply. - **Benchmark-relative** — most funds are measured vs an index; **active weights** (overweight/underweight vs the benchmark) express views; **tracking error** measures deviation. - **Constraints** — position limits, sector caps, liquidity, mandate.

Risk-adjusted performance metrics: | Metric | Formula | Measures | |---|---|---| | **Sharpe ratio** | (Rp − Rf) / σp | Excess return per unit of total risk | | **Treynor ratio** | (Rp − Rf) / βp | Excess return per unit of market risk | | **Alpha (Jensen's)** | Rp − [Rf + β(Rm − Rf)] | Return above CAPM (skill) | | **Information ratio** | Active return / tracking error | Consistency of outperformance vs benchmark | | **Beta** | Cov(Rp,Rm)/Var(Rm) | Sensitivity to the market |

Sharpe uses **total** risk (σ), Treynor uses **systematic** risk (β) — use Treynor when the portfolio is one part of a diversified whole. **Alpha** is the holy grail: return not explained by market exposure.

Rebalancing. Periodically returning to target weights (as prices drift) enforces "sell high, buy low" and controls risk.

Worked examples

Example 1 — diversification cuts risk. Two stocks each with 20% volatility and a correlation of 0.3. A 50/50 portfolio has volatility = $\sqrt{(0.5^2 \cdot 0.2^2 + 0.5^2 \cdot 0.2^2 + 2 \cdot 0.5 \cdot 0.5 \cdot 0.3 \cdot 0.2 \cdot 0.2)} = \sqrt{(0.01 + 0.006)} \approx \sqrt{0.016} \approx$ **12.6%** — well below the 20% of either stock alone, for the average return. That's the diversification benefit from correlation < 1.

Example 2 — Sharpe ratio comparison. Portfolio A returns 15% with 10% volatility; Portfolio B returns 20% with 25% volatility; risk-free 5%. Sharpe A = (15−5)/10 = **1.0**; Sharpe B = (20−5)/25 = **0.6**. B has the higher return but A is the better *risk-adjusted* performer — you'd prefer A (and could lever it to match B's return at lower risk).

Example 3 — alpha vs beta. A fund returned 18% when the market (beta 1.2 exposure) returned 12%, risk-free 5%. CAPM-expected = 5 + 1.2(12−5) = 13.4%. **Alpha = 18 − 13.4 = +4.6%** — genuine outperformance beyond what its market exposure explains. Distinguishing this skill (alpha) from just taking market risk (beta) is the whole game.

How it is tested in interviews

- **"Why diversify?"** — "Because assets aren't perfectly correlated, combining them lowers portfolio volatility for a given return — it removes diversifiable, idiosyncratic risk, leaving mostly systematic risk."
- **"Systematic vs unsystematic risk?"** — "Unsystematic is company-specific and diversifiable; systematic is market-wide, undiversifiable, and measured by beta — it's what earns the risk premium."
- **"What does the Sharpe ratio measure?"** — "Excess return per unit of total risk (volatility) — a risk-adjusted performance measure; higher is better."

- **"Sharpe vs Treynor?"** — "Sharpe uses total risk (σ); Treynor uses systematic risk (β). Use Treynor when the portfolio is part of a broader diversified portfolio."
- **"What is alpha?"** — "Return above what CAPM/beta predicts — the manager's skill, not explained by market exposure."

Traps & common mistakes

- Judging on **raw return** instead of **risk-adjusted** return.
- Thinking diversification removes **all** risk — systematic risk remains.
- Confusing **Sharpe (total risk)** and **Treynor (systematic risk)**.
- Confusing **alpha** (skill) with **beta** (market exposure) — high returns from leverage/beta aren't alpha.
- Over-diversifying to closet-indexing (paying active fees for index-like returns).

First-principles recap

- Portfolios exploit **diversification** (correlations < 1) to raise return per unit of risk.
- Risk = **systematic** (market, priced via beta) + **unsystematic** (specific, diversifiable).
- Judge performance **risk-adjusted**: Sharpe, Treynor, alpha, information ratio.
- **Alpha** = skill beyond market exposure; **beta** = market exposure.
- Rebalance to targets; manage tracking error vs the benchmark.

Quick-reference

Metric	Formula
Beta	Cov(Rp,Rm)/Var(Rm)
Sharpe	(Rp − Rf)/σp
Treynor	(Rp − Rf)/βp
Alpha	Rp − [Rf + β(Rm − Rf)]
Information ratio	Active return / tracking error
Diversifiable	Unsystematic only

ESG in Equity Analysis

The Problem / Why this matters

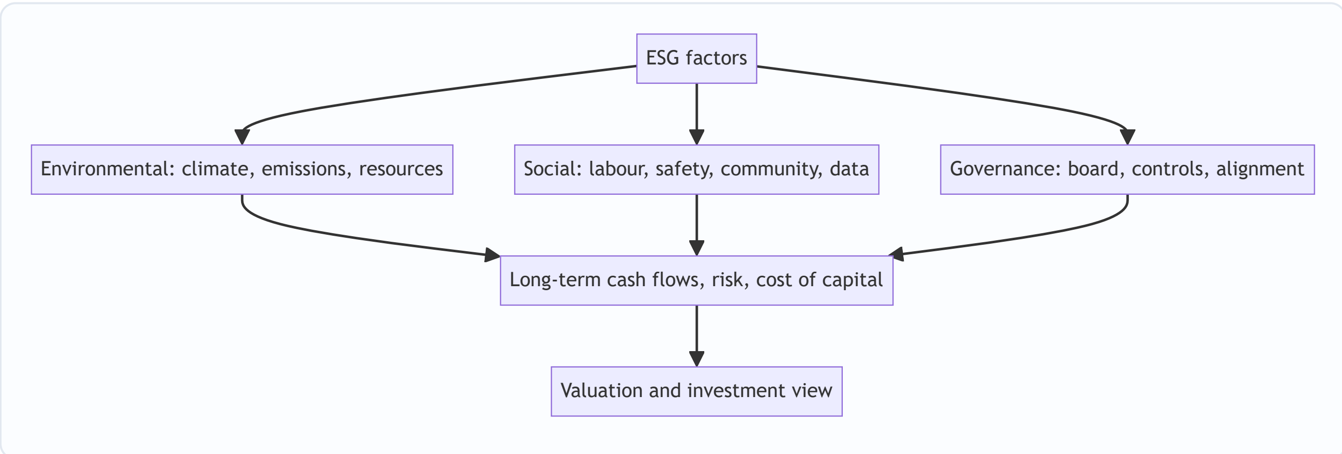
Investors increasingly judge companies not just on financials but on **Environmental, Social and Governance (ESG)** factors — because these non-financial issues create real financial risks and opportunities (regulation, reputation, stranded assets, talent). ESG has moved from niche to mainstream, is increasingly mandated in disclosure (BRSR in India), and comes up in interviews as "why do investors care about ESG?" Understanding how to integrate it into equity analysis is now part of the job.

Core Idea

ESG analysis assesses a company's exposure to and management of **environmental, social, and governance** risks and opportunities, integrating them into the investment view. The premise: material ESG factors affect long-term cash flows, risk, and therefore valuation — so ignoring them means missing risks that standard financials don't capture.

Why it works this way

Many ESG issues are **financially material but slow-moving** — a carbon-intensive business faces future regulation and stranded-asset risk; poor governance invites fraud and value destruction; weak labour or safety practices bring litigation and reputational damage. These show up in cash flows and cost of capital eventually, so a forward-looking analyst prices them in before they hit the financials.



Full technical content

The three pillars: | Pillar | Example factors | |---|---| | **Environmental** | Carbon emissions, energy/water use, waste, climate transition risk, biodiversity | | **Social** | Labour practices, health & safety, diversity, product safety, data privacy, community relations | | **Governance** | Board independence/quality, executive pay alignment, shareholder rights, audit quality, related-party dealings, anti-corruption |

Governance is often seen as the most immediately financially material — weak governance (dominant promoters, poor controls) is a leading cause of fraud and value destruction, especially in emerging markets.

How investors use ESG: - **Integration** — embed material ESG factors into fundamental analysis and valuation (adjust growth, margins, cost of capital, or apply a risk discount). - **Screening** — exclude (negative screening: tobacco, weapons) or tilt toward leaders (positive/best-in-class). - **Thematic / impact** — invest in a theme (clean energy) or for measurable impact. - **Stewardship** — engage and vote to improve company behaviour.

Materiality. The key discipline: focus on ESG factors that are **financially material** to *that* industry (e.g., emissions for a utility, data privacy for a tech firm, safety for a miner) — not a generic checklist. Frameworks like SASB map materiality by sector.

Instruments & disclosure: green bonds, **sustainability-linked loans**, ESG funds/ETFs; disclosure via GRI, TCFD, and in India the **BRSR (Business Responsibility and Sustainability Report)** mandated for top listed firms by SEBI.

Debates & challenges: - **Greenwashing** — overstating ESG credentials. - **Inconsistent ratings** — ESG scores from different providers often disagree (different methodologies). - **Performance debate** — whether ESG helps or hurts returns is contested; the strongest case is **risk management** (avoiding blow-ups) rather than guaranteed outperformance.

Effect on cost of capital. Strong ESG can lower cost of capital (broader investor base, lower risk perception); poor ESG can raise it (exclusion, higher risk), feeding directly into valuation.

Worked examples

Example 1 — governance red flag. An analyst finds a company with a promoter-dominated board, frequent related-party transactions, and an auditor change after a qualification. Governance risk is high — historically a precursor to value destruction. The analyst applies a valuation discount (or avoids the stock), a call standard financials alone wouldn't trigger.

Example 2 — environmental transition risk. A thermal-coal power producer looks cheap on current earnings, but tightening emissions regulation and the energy transition threaten **stranded assets** and rising compliance costs over the next decade. Integrating this, the analyst lowers long-term cash flows and raises the discount rate — the stock is less cheap than it looks.

Example 3 — materiality focus. For a software company, carbon emissions are minor but **data privacy and talent retention** are highly material (a breach or attrition could hit revenue and margins). A good ESG analyst weights the *material* factors for that sector, not a generic environmental score.

How it is tested in interviews

- **"Why do investors care about ESG?"** — "Because material ESG factors create real financial risks and opportunities — regulation, stranded assets, reputation, governance failures — that affect long-term cash flows and cost of capital but aren't captured in standard financials."
- **"What are the three pillars?"** — Environmental (climate, emissions, resources), Social (labour, safety, privacy, community), Governance (board, controls, alignment).
- **"Which is most financially material?"** — "Often governance — weak governance is a leading cause of fraud and value destruction, especially in emerging markets — but materiality is industry-specific."
- **"What are the criticisms of ESG?"** — Greenwashing, inconsistent ratings across providers, and a contested performance record (the strongest case is risk management).
- **"How would you integrate ESG into a valuation?"** — "Focus on material factors for the sector; adjust growth, margins, or the discount rate, or apply a governance/risk discount."

Traps & common mistakes

- Treating ESG as a **generic checklist** rather than **industry-material** factors.
- Ignoring **governance** — the most immediately material pillar.
- Taking a single ESG **rating** at face value (providers disagree).
- Assuming ESG **guarantees** outperformance — the robust case is risk management.
- Missing the **cost-of-capital** channel through which ESG hits valuation.

First-principles recap

- ESG assesses environmental, social and governance risks/opportunities that are financially material.
- **Materiality is industry-specific** — focus on what matters for that sector.
- Governance is often the most immediately material pillar.
- Integrate via adjusted cash flows/discount rate, screening, or stewardship.
- Watch greenwashing and rating inconsistency; the strongest case is risk management and cost of capital.

Quick-reference

Pillar	Material examples
Environmental	Emissions, transition risk, resources
Social	Labour, safety, data privacy
Governance	Board, controls, related-party, alignment
Uses	Integration, screening, thematic, stewardship
India disclosure	BRSR (SEBI)
Key discipline	Focus on financially material factors

Indian Equity Markets (Capstone)

The Problem / Why this matters

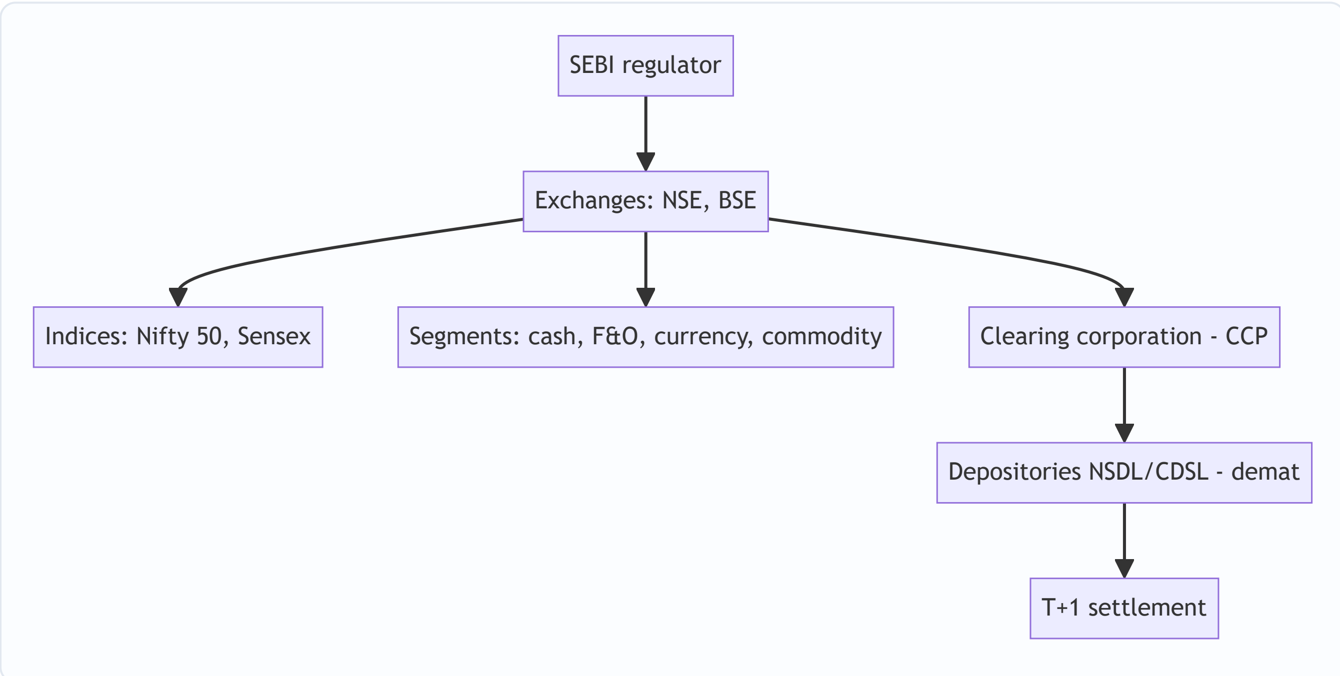
For any finance role in India, you must know your home market cold — the regulator, the exchanges, the indices, how shares are held and settled, the segments, and the participants. It's the most common "home turf" interview check, and it ties together everything in this book in the specific institutional context you'll actually work in. This capstone maps the Indian equity ecosystem end to end.

Core Idea

The Indian equity market is a **modern, electronic, well-regulated** market: **SEBI** regulates it; **NSE and BSE** are the exchanges; **Nifty 50 and Sensex** are the benchmarks; shares are held in **demat** via NSDL/CDSL and settle at **T+1**; and it spans cash, F&O, currency and commodity segments with deep participation from retail, domestic institutions, and foreign investors.

Why it works this way

India built a fully dematerialized, exchange-traded, centrally-cleared market with strong disclosure and one of the world's fastest settlement cycles precisely to ensure trust, liquidity and investor protection — the conditions that let a large, diverse investor base participate and companies raise capital efficiently.



Full technical content

Regulator: SEBI (Securities and Exchange Board of India) — regulates issuers (disclosure), intermediaries (registration/conduct), and markets (surveillance, insider-trading and manipulation enforcement); protects investors. **RBI** oversees monetary policy and banking; **IRDAI** insurance.

Exchanges: NSE (National Stock Exchange — the larger by volume, especially derivatives) and **BSE** (Bombay Stock Exchange — Asia's oldest). Both are fully electronic, order-driven markets.

Benchmark indices: Nifty 50 (NSE, 50 large-caps) and **Sensex** (BSE, 30 large-caps), both **free-float market-cap weighted**. Broader: Nifty Next 50, Nifty 100/500, Midcap/Smallcap, and sectoral indices (**Bank Nifty**, Nifty IT, etc.).

Holding & settlement: shares held in **dematerialized (demat)** form via depositories **NSDL** and **CDSL** (through Depository Participants); trades cleared by the **clearing corporation** (central counterparty) and settled on **T+1** (India moved to T+1 fully in 2023 and is piloting T+0/instant) — among the fastest globally.

Market segments: | Segment | What trades | |---|---| | **Cash / equity** | Delivery-based and intraday shares | | **F&O (derivatives)** | Index and stock futures & options (Nifty, Bank Nifty are among the world's most traded) | | **Currency derivatives** | USD/INR and other pairs | | **Commodity** | Via MCX/NCDEX (and NSE/BSE segments) |

Participants: retail (large and growing, via demat/UPI-based investing), **domestic institutions (DIIs)** — mutual funds, insurers (LIC), pension (EPFO/NPS) — and **foreign portfolio investors (FPIs)**, whose flows often drive index moves. Promoters hold large stakes in many firms.

Trading mechanics: order-driven, price-time priority; **circuit breakers** (index-level halts) and **price bands** (stock-level) curb extreme moves; **UPI/ASBA** streamline IPO applications and payments.

Recent evolution: surging retail participation (record demat accounts), the rise of discount brokers and index/SIP investing, T+1 (moving to instant) settlement, and world-leading F&O volumes (prompting SEBI curbs on retail options speculation).

Worked examples

Example 1 — the full trade lifecycle. A retail investor places a market buy for 100 shares of an NSE-listed company via a broker app. The order matches on the NSE by price-time priority; the **clearing corporation** guarantees it; on **T+1**, shares are credited to the investor's **CDSL/NSDL demat** account and cash debited. Regulated end-to-end by SEBI. Every institution in this chapter touched one small trade.

Example 2 — FPI flows move the Nifty. Foreign investors turn net sellers of ₹15,000 cr in a week on global risk-off. Because FPIs hold large weights in Nifty heavyweights and trade fast, the index falls sharply even without India-specific news — illustrating why FPI flows are watched as a key driver.

Example 3 — F&O depth. Bank Nifty and Nifty options are among the most actively traded derivatives in the world, giving Indian markets deep hedging and speculation venues — but also prompting SEBI to tighten rules to protect retail traders from outsized options losses. India's derivatives market is globally significant.

How it is tested in interviews

- **"Walk me through the Indian equity market structure."** — SEBI regulates; NSE/BSE are the exchanges; Nifty 50/Sensex are the benchmarks (free-float market-cap weighted); shares held in demat via NSDL/CDSL; cleared by the clearing corporation; settled T+1; segments cash/F&O/currency/commodity.
- **"Who regulates the Indian securities market?"** — "SEBI — issuers, intermediaries, and markets, with investor protection; RBI covers banking/monetary, IRDAI insurance."
- **"What are the main indices and how are they built?"** — "Nifty 50 (NSE) and Sensex (BSE), free-float market-cap weighted large-cap baskets."
- **"How are trades settled in India?"** — "T+1 rolling settlement, shares in demat via NSDL/CDSL, guaranteed by the clearing corporation as central counterparty — among the fastest cycles globally."

- **"What drives Indian market moves?"** — "FPI and DII flows, global cues, rates/inflation and RBI policy, earnings, and sentiment; FPI flows especially move the index."

Traps & common mistakes

- Confusing **SEBI** (securities) with **RBI** (banking/monetary) roles.
- Not knowing India settles at **T+1** (and is going faster).
- Forgetting Nifty/Sensex are **free-float** market-cap weighted.
- Mixing up the **exchange** (NSE/BSE), **depository** (NSDL/CDSL), and **clearing corporation** roles.
- Underrating **FPI flows** and India's globally-large **F&O** volumes.

First-principles recap

- **SEBI** regulates; **NSE/BSE** are the exchanges; **Nifty 50/Sensex** the free-float benchmarks.
- Shares are held in **demat** (NSDL/CDSL) and settle **T+1** (moving to instant).
- Segments: cash, F&O (globally large), currency, commodity.
- Participants: retail (surging), DIIs, FPIs (flow-drivers), promoters.
- A modern, electronic, centrally-cleared, well-regulated market — the context for every Indian finance role.

Quick-reference

Element	India
Regulator	SEBI (RBI banking, IRDAI insurance)
Exchanges	NSE, BSE
Indices	Nifty 50, Sensex (free-float mcap)
Depositories	NSDL, CDSL (demat)
Settlement	T+1 (→ instant), CCP-guaranteed
Segments	Cash, F&O, currency, commodity
Flow drivers	FPIs, DIIs